

# Network IP Camera Application Programming Interface (NIPCA)

Ver. 1.9.5

## Document History

| Version | Date       | Comment                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|---------|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0.99a   | 2007-11-09 | Focus on the configuration settings.                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 0.99b   | 2007-11-20 | Add the Valid values. Add RS-485 commands.<br>Refine all document.                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 0.99c   | 2007-11-21 | Modify HTTP status codes, basic info, datetime DST, motion detection.                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 0.99d   | 2007-12-14 | 2.3: Modify HTTP status codes description.<br>Add descriptions and examples of parameters and values.<br>3.3.4 3.3.5: Modify flicker, autoexposure for sensor_info.cgi sensor.cgi.<br>3.4.8: Modify upnpav, upnpcp for upnp.cgi.<br>5.1.3: Remove redundant penable, tenable, zenable for ptz_step.cgi.<br>5.1.5: Add p,t,z position for ptz_preset_list.cgi.<br>6.1.2: Add keep_alive for notify_stream.cgi.                                                                            |
| 0.99e   | 2007-12-26 | Add the 3.5 event handling.                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 1.0     | 2008-01-14 | 3.1.3 Modify the request to /users/verify.cgi<br>3.1.6 3.1.7 Modify method and offset.<br>Add the ID of the table dynamic DNS service providers.                                                                                                                                                                                                                                                                                                                                         |
| 1.1     | 2008-01-31 | 2.1 Fix the POST Content-Type to application/x-www-form-urlencoded.<br>Add the ACS Stream Header.                                                                                                                                                                                                                                                                                                                                                                                        |
| 1.2     | 2008-02-21 | 2.1 Fix the POST parameters.                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 1.3     | 2008-05-26 | 3.5.1 change the definition of macro block size.<br>3.3.4 add hue, autoexposure, autogainctrl<br>3.3.5 add hue, autogainctrl<br>3.6.8 add delaytime                                                                                                                                                                                                                                                                                                                                      |
| 1.4     | 2008-05-16 | 3.1.1 get basic information: add 'nipca' item<br>3.1.1 add 'videoout' item<br>3.3.4 add 'inputsize' and 'videooutformat' item.<br>3.3.5 add 'color' item.<br>3.5.1 add more actions like cifs_rec, cifs_shot...<br>3.5.4 add more actions and the field prerecord and postrecord.<br>3.5.15, 3.5.16 add keep space item.<br>4.1.7 add get video stream of associated profile<br>6.1.1, 6.1.2 add mdv# item<br>7.1 change url of rtsp:<br>mpeg4 -> mp4<br>mjpeg -> jpeg<br>add /live# url |

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| 1.5   | 2008-10-17 | <p>3.3.1, 3.3.2 add vprofileformat for video stream cgi 4.1.7</p> <p>3.5.3, 3.5.4 add sw_input event</p> <p>3.5.3, 3.5.4 change ‘actions’ and ‘action’ keyword to be ‘handlers’ and ‘handler’</p> <p>5.3 add an software event trigger function</p> <p>4.1.5, 4.1.6 revise the part of format description</p> <p>5.2.1 unify the speed range to 1-10</p> <p>4.1.8 add put audio upstream (two-way audio talk)</p> <p>4.1.9 add H.264 streaming cgi url</p> <p>Add 7.1.1-7.1.3 to support customized url entry of RTSP live stream.</p> <p>3.4.1 add httpexternalport, rtspport and rtspexternalport.</p>                                                                                                                                                                                                                                                                                             |
| 1.6   | 2009-02-17 | <p>5.3.1 add ‘trigger’ item to indicate that client want to turn this event on or off.</p> <p>3.3.1, 3.3.2, 4.1.7 Refine the definition of vprofileformat</p> <p>5.1.1 add item “customizedhome” to indicate whether camera support 5.1.9 function or not.</p> <p>5.1.9 add PTZ home manifest command.</p> <p>3.1.13 add ‘reset sensor to default configuration’ function</p> <p>3.3.5, 3.3.6 add videoinformat for some video server models</p> <p>3.3.14, 3.3.15 add IR LED setting functions</p> <p>3.3.16, 3.3.17 add ICR setting functions</p>                                                                                                                                                                                                                                                                                                                                                  |
| 1.7   | 2009-06-10 | <p>3.3.18, 3.3.19 add authentication control for live video and snapshot</p> <p>3.1.1 add field ‘product’, ‘brand’ to basic information</p> <p>5.1.10, 5.1.11 add Auto Patrol/Auto Pan for PTZ control function</p> <p>5.1.12 Configure autot patrol preset sequence order.</p> <p>3.6.5 add example html to submit firmware to ip camera.</p> <p>4.1.2 add ‘profileid’ optional parameter</p> <p>4.1.3 change this section to MPEG-4 elementary stream CGI.</p> <p>4.1.5, 4.1.6, 4.1.9 refine the description of parameter ‘profileid’</p> <p>4.1.7 modify the format of stream with profile M-JPEG (replaced by ACVS wrapped stream)</p> <p>4.1.11 add audio profile stream CGI</p> <p>8.2 Add two more frame type for ACVS header.</p> <p>3.3.4, 3.3.5, 3.3.6 add sharpness</p> <p>3.4.14, 3.5.15 add wireless strength function and wireless site survey</p> <p>3.7 SD card operations added</p> |
| 1.8   | 2009-09-30 | <p>4.1.8 Modify audio uploading method</p> <p>3.7 Update whole SD card section.</p> <p>3.5.11 Add enable, prefix and interval field.</p> <p>3.5.13 Add enable field.</p> <p>3.5.15 Add enable field.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 1.9   | 2010-4-22  | <p>5.1.13 get, set the type of focus function: auto focus or manual focus</p> <p>5.1.14 adjust the focus manually, focus in or focus out from current position</p> <p>Remove unsupport list.</p> <p>Fix wording and wrong statements.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 1.9.1 | 2010-7-7   | <p>3.5.3. Newly add get Recorder action</p> <p>3.5.4. Newly add set Recorder action</p> <p>3.5.5. Newly add get Snapshot action</p> <p>3.5.6. Newly add set Snapshot action</p> <p>3.5.7. Newly add get Alarm out action</p> <p>3.5.8. Newly add set alarm out action</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

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| 1.9.2 | 2010-8-4   | <p>3.4.15. Newly add get HTTPS configuration</p> <p>3.4.16. Newly add set HTTPS configuration</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 1.9.3 | 2010-10-31 | <p>3.3.2. 3.3.3.Modify quality not only for MJPEG and Newly add qualitymodes</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 1.9.4 | 2011-7-01  | <p>3.3.2. Newly add get sensor output configuration</p> <p>3.3.3. Newly add set sensor output configuration</p> <p>3.3.1. Modify the resolutions and vprofileres# description.</p> <p>3.3.4. Modify the resolution description.</p> <p>3.3.6. Add flicker, mirror, flip, color</p> <p>5. Add /ptz/ directory to PTZ privilege group.</p> <p>5.1.15., 5.1.16. Add get/set PTZ privilege group.</p> <p>3.3.1. Add parameters: cur_micvol, cur_speakervol</p> <p>5.3. PTDC Pan/Tilt get information</p> <p>5.4. PTDC Pan/Tilt set information</p> <p>5.5. PTDC Pan/Tilt other Parts</p> <p>5.6. PTDC Zoom/Focus/Focus Type get information</p> <p>5.7. PTDC Zoom/Focus/Focus Type set information</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 1.9.5 | 2011-11-22 | <p>3.3.4 Newly add get video type.</p> <p>3.3.5 Newly add set video type.</p> <p>3.3.18 Newly add set IR LED.</p> <p>3.3.19 Newly add get ICR(Infrared Cut filter Removal) settings.</p> <p>3.3.20 Newly add set ICR(Infrared Cut filter Removal) settings.</p> <p>3.3.23 Newly add query Privacy mask information</p> <p>3.3.24 Newly add get Privacy mask</p> <p>3.3.25 Newly add set Privacy mask</p> <p>3.4.17 Newly add list all the IP access list.</p> <p>3.4.18 Newly add add, delete acces IP.</p> <p>3.6.10 Newly add get Privacy mode configuration.</p> <p>3.6.11 Newly add set Privacy mode configuration.</p> <p>3.6.12 Newly add get TV output.</p> <p>3.6.13 Newly add eet TV output.</p> <p>3.6.14 Newly add get DC power.</p> <p>3.6.15 Newly add set DC power.</p> <p>3.6.16 Newly add get device timestamp.</p> <p>3.6.17 Newly add set device timestamp.</p> <p>4.1.10 Newly add get audio MS-ADPCM stream</p> <p>4.1.11 Newly add get audio MU-LAW stream</p> <p>4.1.12 Newly add get audio AAC stream</p> <p>4.1.13 Newly add get audio A-LAW stream</p> <p>5.1.17 Newly add query focus information.</p> <p>5.1.18 Newly add get the current position of focus.</p> <p>5.1.19 Newly add set absolutely position of focus.</p> <p>5.1.20 Newly add fine-tune focus automatically.</p> <p>5.1.21 Newly add PTZ direction of movement.</p> <p>5.8.1 Newly add get the current digital PTZ position.</p> <p>5.8.2 Newly add add, delete or goto a digital PTZ preset.</p> <p>5.8.3 Newly add move digital PTZ absolutely.</p> <p>5.8.4 Newly add move digital PTZ relatively.</p> <p>5.8.5 Newly add digital PTZ autopan.</p> <p>5.8.6 Newly add digital PTZ sequence.</p> <p>3.1.1 Add field “focus” “pir” “irc” and “ir” to basic information.</p> <p>3.3.1 Add “resolutionlist#”, “frameratelist#” fields.</p> <p>3.3.6, 3.3.7 Add “viewwindow” fields.</p> |

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|  |  | <p>3.3.8 Add “wds” “exposuremode” , “maxshutter” , “minshutter” . “maxgain” , “noisereduction” , “wdrlevel” fields.</p> <p>3.3.9, 3.3.10 Add “wds” “exposuremode” , “maxshutter” , “minshutter” . “maxgain” , “noisereduction” , “wdrlevel” , “wdrlevel” fields.</p> <p>3.3.11, 3.3.12 Add audio codec format value: MU-LAW, A-LAW.</p> <p>3.5.1, 3.5.2 Add “percentage” “pir” fields.</p> <p>3.5.4 Add “fileLenMin” field for setting recording file size by minute(s).</p> <p>3.7.1 Add status values.</p> <p>3.7.2 Specify response format.</p> <p>6.1.1, 6.1.2 Add “irled” “autofocusbusy” field.</p> <p>8.2 Add audio data format value of ACS audio header to support AAC and A-LAW audio codec format.</p> <p>3.5.3, 3.5.4, 3.5.5 Errata correction.</p> <p>Refine TimeZone and Day Light Saving Time as below</p> <ol style="list-style-type: none"> <li>1. TimeZone2: from “Midway Island, Samoa” to “Samoa”</li> <li>2. TimeZone14: from “Bogota, Lima, Quito, Rio Branco” to “Bogota, Lima, Quito”</li> <li>3. TimeZone17: from “Caracas, La Paz” to “La Paz, Georgetown”</li> <li>4. TimeZone22: from “Buenos Aires, Georgetown” to “Buenos Aires”</li> <li>5. TimeZone30: from “Casablanca, Monrovia, Reykjavik” to “Monrovia, Reykjavik”</li> <li>6. TimeZone41: from “GMT+02:00” to “GMT+03:00”</li> <li>7. TimeZone43: from “GMT+02:00” to “GMT+01:00”</li> <li>8. TimeZone47: from “GMT+03:00” to “GMT+04:00”</li> <li>9. TimeZone55: from “GMT+05:00 Ekaterinburg” to “GMT+06:00 Yekaterinburg”</li> <li>10. TimeZone61: from “(GMT+06:00) Almaty, Novosibirsk” to “(GMT+07:00) Novosibirsk”</li> <li>11. TimeZone63: from “GMT+07:00” to “GMT+08:00”</li> <li>12. TimeZone67: from “GMT+08:00 Irkutsk, Ulaan Bataar” to “GMT+09:00 Irkutsk”</li> <li>13. TimeZone70: from “GMT+09:00” to “GMT+10:00”</li> <li>14. TimeZone80: from “(GMT+11:00) Magadan, Solomon Is., New Caledonia” to “(GMT+12:00) Magadan”</li> <li>15. TimeZone81: from “Fiji, Kamchatka, Marshall Is.” to “Fiji”</li> <li>16. TimeZone83: from “Nuku'alofa” to “Nukualofa”</li> <li>17. TimeZone84: newly add “(GMT-04:30) Caracas”</li> <li>18. TimeZone85: newly add “(GMT+11:00) Solomon Is., New Caledonia”</li> <li>19. TimeZone86: newly add “(GMT) Casablanca”</li> <li>20. TimeZone87: newly add “(GMT+08:00) Ulaanbaatar”</li> </ol> |
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# 1.Overview

---

Network IP Camera Access Application Programming Interface (NIPCA-API) is a HTTP-based API for the networks IP camera products. Users can write program easily by calling this API to access all functionalities provided by our IP cameras including: configuration, multimedia streaming data and the control facilities.

Except Streaming, the other groups of API use the same format in transporting HTTP-based message. We will describe the command HTTP request format in the next chapter.

As for the Streaming API, the output format of streaming API depends on different IP camera model. Here we only provide a general entry point to let the IP camera outputting streaming via a permanent HTTP connection.

We also provide the RTSP interface for our IP camera.

## 1.1.API versions

Though we provide a common API for all IP camera models, it may not apply to some old models which were produced before the first version of this API being published. We may also publish the further version of this API in the future. So there may be some difference between different versions of API. However, all our products shall provide the API version information with every firmware version of each model.

## 1.2.Valid values

The following valid values are used in this document:

| Values                               | Description                                                                                                                               |
|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| An integer                           | Any number between -2147483647 ( $-2^{31}-1$ ) and 2147483647 ( $2^{31}-1$ ).                                                             |
| m ... n                              | Any number between number m and number n.                                                                                                 |
| #                                    | A number equals or greater than 0                                                                                                         |
| A string                             | Any string encoded by UTF-8                                                                                                               |
| An IP address                        | A string limited to contain an IP address of the format xxx.xxx.xxx.xxx, where xxx is a number between 0 to 255.<br>Example: 192.168.0.90 |
| A MAC Address                        | A string limited to contain a MAC address of the format xx:xx:xx:xx:xx:xx, where xx is a hexadecimal value.<br>Example: 00:40:8c:cd:00:00 |
| A time                               | A string limited to contain a time of the format hh:mm:ss.<br>Example: 23:01:14                                                           |
| A date                               | A string limited to contain a date of the format yyyy-mm-dd.<br>Example: 2004-02-16                                                       |
| <value 1>, <value 2>, <value 3>, ... | Enumeration, only the given values are valid.                                                                                             |
| < <i>italic string</i> >             | Every italic strings inside brackets including the brackets should be replaced by proper values.                                          |

## 2.HTTP Interface

---

An HTTP-based protocol always includes two kinds of message, request and response. IP camera prepares a service to wait and accept TCP connection request with a specified port and to process the requests message from a user defined application. In this chapter, we will describe the common format of comprising all the different request and response messages. Although our camera also can support HTTP/1.0, we recommend that a request compliant with HTTP/1.1 may encounter fewer problems. You may also refer to the RFC 2616 HTTP/1.1.

### 2.1.Request messages

To query information of IP camera, use the syntax

```
GET http://<camera name>/<CGI-URL>?<parameter>=<value> HTTP/1.1<CRLF>
```

```
Authorization: Basic <basic-cookie><CRLF>
```

```
Host: <camera ip-adress><CRLF>
```

```
<CRLF>
```

where,

<CGI-URL> is a URL of a CGI. For example, get basic information is “/common/info.cgi”.

Authorization is optional for some CGIs.

<basic-cookie> is the base64 encoding of userid:password. (Notes: For some models, MD5 DIGET may be used.)

<CRLF> is Carriage Return and Line Feed (\r\n) .

To set values in the IP camera, you may use HTTP GET method, the syntax is

```
GET http://<camera name>/<CGI-URL>
```

```
?<parameter>=<value>[&<parameter>=<value>...] HTTP/1.1<CRLF>
```

```
Authorization: Basic <basic-cookie><CRLF>
```

```
Host: <camera ip-adress><CRLF>
```

```
<CRLF>
```

or HTTP method POST, the syntax is

```
POST http://<camera name>/<CGI-URL> HTTP/1.1<CRLF>
```

```
Authorization: Basic <basic-cookie><CRLF>
```

```
Host: <camera ip-adress><CRLF>
```

```
Content-Type: application/x-www-form-urlencoded<CRLF>
```

```
Content-Length: <body length><CRLF>
```

```
<CRLF>
```

```
<parameter>=<value>[&<parameter>=<value>]
```

where,

<body length> is the length of the entity body.

<parameter> will be described in the following chapters. Valid characters only include alphabets([A-Za-z]), digits([0-9]) and underline(\_). There is no such restriction for <value>. The content part of the post message should be encoded with “url-encoding” function.

## 2.2.Response messages

While IP Camera receives request message from user, it will do the related action then output result as response message:

```
HTTP/1.1 <HTTP code> <HTTP text><CRLF>
```

```
Content-Type: text/plain<CRLF>
```

```
Content-Length: <body length><CRLF>
```

```
<CRLF>
```

```
<parameter>=<values><CRLF>
```

```
...
```

## 2.3.Response status codes

The API status codes are defined here.

*Table 1: HTTP status codes*

| HTTP code | HTTP text           | Description                                                                                       |
|-----------|---------------------|---------------------------------------------------------------------------------------------------|
| 200       | OK                  | The request has succeeded, but an application error may occur, please refer to each CGI response. |
| 400       | Bad Request         | You used invalid or unsupported parameters or values for this IP camera.                          |
| 401       | Unauthorized        | The request requires user authentication or the authorization was refused.                        |
| 404       | Not Found           | This API is not supported for this IP camera.                                                     |
| 500       | Internal Error      | The IP camera encountered an internal error or the API can not get the correct status.            |
| 503       | Service Unavailable | The IP camera is unable to handle the request due to temporary overload.                          |

## 3.Configuration API

The CGIs under /config can only be accessed by administrators. Most of CGIs are one-shot commands, which only return current configurations and status of IP camera and terminated. If you need to monitor camera status for a long time, please use 6.1.2. notify\_stream.cgi instead.

### 3.1.device information

#### 3.1.1.get basic information

request:

GET /common/info.cgi

No authentication required.

response:

| Name     | Value         | Description                                                                                             |
|----------|---------------|---------------------------------------------------------------------------------------------------------|
| model    | A string      | model name                                                                                              |
| product  | A string      | product name of camera                                                                                  |
| brand    | A string      | brand name                                                                                              |
| version  | A string      | version number of firmware                                                                              |
| build    | A string      | firmware build number                                                                                   |
| nipca    | A string      | version number of NIPCA supported (e.g. 1.2, 1.4)                                                       |
| name     | A string      | camera name                                                                                             |
| location | A string      | camera location                                                                                         |
| macaddr  | A MAC address | the MAC address                                                                                         |
| ipaddr   | An IP address | IP address of current active network interface. (Notes, this will not be a IP address of PPPoE.)        |
| netmask  | An IP address | Subnet mask                                                                                             |
| gateway  | An IP address | Default router/gateway used for connecting devices attached to different networks and networks segment. |
| wireless | yes, no       | Or omitted if the IP camera doesn't have a wireless.                                                    |
| ptz      | P, T, Z       | Only show supported Pan or Tilt or Zoom.<br>For example, ptz=P,T                                        |
| focus    | yes, no       | Or omitted if the IP camera doesn't have a focus function.                                              |
| inputs   | #             | The number of AlarmIN.                                                                                  |
| outputs  | #             | The number of AlarmOUT.                                                                                 |
| speaker  | yes, no       | Or omitted if the IP camera doesn't have a speaker.                                                     |
| Videoout | yes, no       | Or omitted if the IP camera doesn't have video out.                                                     |
| pir      | yes, no       | Or omitted if the IP camera doesn't have PIR.                                                           |
| icr      | yes, no       | Or omitted if the IP camera doesn't have ICR                                                            |
| ir       | yes, no       | Or omitted if the IP camera doesn't have IR                                                             |

#### 3.1.2.quickly verify user

request:

GET /users/verify.cgi

response:

| Name  | Value    | Description                                                                  |
|-------|----------|------------------------------------------------------------------------------|
| group | A string | the group name of the specified user in the HTTP Authorization header field. |

When the authorization fails, it will return HTTP/1.0 401 Unauthorized

### 3.1.3.get camera info

request:

GET /config/camera\_info.cgi

response:

| Name     | Value    | Description     |
|----------|----------|-----------------|
| name     | A string | camera name     |
| location | A string | camera location |

### 3.1.4.set camera info

request:

GET/POST /config/camera\_info.cgi

parameters:

see the above table.

response:

see the above table.

### 3.1.5.get system date and time

request:

GET /config/datetime.cgi

response:

| Name       | Value                | Description                                                                                                                                        |
|------------|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| method     | 0, 1                 | 0: disable ntpd<br>1: enable ntpd                                                                                                                  |
| timeserver | A host or IP address | NTP time server host name or IP address.                                                                                                           |
| timezone   | #                    | time zone ID, see Table <a href="#">Time zone</a>                                                                                                  |
| date       | A date               | yyyy-mm-dd                                                                                                                                         |
| time       | A time               | hh:mm:ss                                                                                                                                           |
| dstenable  | no, yes              | disable or enable the DST (Daylight Saving Time)                                                                                                   |
| dstauto    | no, yes              | set DST automatically                                                                                                                              |
| offset     | A time               | The amount of time the clock should be turned back/forward (hh:mm), due to DST.                                                                    |
| starttime  |                      | The time when DST should be enabled in the format m.w.d/hh:mm:ss<br>day d (0 ... 6) of week w (1 ... 5) of month m (1 ... 12).<br>d=0 is a Sunday. |
| stoptime   |                      | Stop time when DST should be disabled in the same format as above.                                                                                 |

### 3.1.6.set system date and time

request:

GET/POST /config/datetime.cgi

parameters:

| Name       | Value                | Description                                                                                                                    |
|------------|----------------------|--------------------------------------------------------------------------------------------------------------------------------|
| method     | 0, 1, 2              | 0: disable ntpd<br>1: enable ntpd<br>2: manual setting, requires date and time.                                                |
| timeserver | A host or IP address | NTP time server host name or IP address.                                                                                       |
| timezone   | 1 ... 83             | time zone ID, see Table <a href="#">Time zone</a>                                                                              |
| date       | A date               | yyyy-mm-dd                                                                                                                     |
| time       | A time               | hh:mm:ss                                                                                                                       |
| dstenable  | no, yes              | disable or enable the DST (Daylight Saving Time)                                                                               |
| dstauto    | no, yes              | set DST automatically                                                                                                          |
| offset     | A time               | The amount of time the clock should be turned back/forward (hh:mm), due to DST.                                                |
| starttime  |                      | The time when DST should be enabled in the format m.w.d/hh:mm:ss<br>day d (0 ... 6) of week w (1 ... 5) of month m (1 ... 12). |

|          |  |                                                                    |
|----------|--|--------------------------------------------------------------------|
|          |  | d=0 is a Sunday.                                                   |
| stoptime |  | Stop time when DST should be disabled in the same format as above. |

response:  
see the 3.1.6 table.

## 3.2.users and groups

### 3.2.1.get users

request:  
GET /config/user\_list.cgi

parameters:  
none or  
name=<username>

response:  
if no request parameter

| Name              | Value               | Description                                                                 |
|-------------------|---------------------|-----------------------------------------------------------------------------|
| users             | #                   | The total number of users.                                                  |
| <username><br>... | <group name><br>... | For example, admin=admingrp<br>It will display all user names line by line. |

if request parameter is name

| Name  | Value    | Description                           |
|-------|----------|---------------------------------------|
| group | A string | the group which this user belongs to. |

### 3.2.2.add or modify a user

request:  
GET/POST /config/user\_mod.cgi

parameters:

| Name     | Value    | Description                           |
|----------|----------|---------------------------------------|
| name     | A string | user name                             |
| password | A string | base64 encoded password               |
| group    | A string | the group which this user belongs to. |

response:  
see the above table.

### 3.2.3.delete users

request:  
GET/POST /config/user\_del.cgi

parameters:  
name =<username1>,<username2>, ...  
You can delete many users at once.

response:  
name=<username1>,<username2>, ...

### 3.2.4.get groups

request:

GET /config/group\_list.cgi

parameters:

none or

name=<groupname>

response:

if no request parameter

| Name               | Value               | Description                                                                                 |
|--------------------|---------------------|---------------------------------------------------------------------------------------------|
| groups             |                     | The total number of groups.                                                                 |
| <groupname><br>... | <user1>, ...<br>... | for example, admingrp=admin,root<br>It will display all group names and users line by line. |

if request parameter is name

| Name      | Value                                        | Description                                |
|-----------|----------------------------------------------|--------------------------------------------|
| user      | <user1>, ...                                 | the user names                             |
| privilege | ptz, outputs, speaker, mic, video,<br>notify | the permissions list which this group has. |

## 3.3.video, sensor, audio

### 3.3.1.query stream information

You can get supported parameter values for your IP camera. Some parameters are optional and not displayed if not supported in your IP camera.

request:

GET /config/stream\_info.cgi

GET /users/stream\_info.cgi (accessible by all viewers groups)

response:

| Name            |  | Value                | Description                                                                                                                      |
|-----------------|--|----------------------|----------------------------------------------------------------------------------------------------------------------------------|
| videos          |  | MPEG4, MJPEG, H264   | available video codecs list. for example, videos=MPEG4,MJPEG                                                                     |
| audios          |  | PCM, ADPCM, AMR, AAC | available audio codecs list. for example, audios=PCM                                                                             |
| resolutions     |  | <width>x<height>,... | available video resolutions list. for example, resolutions=640x480,320x240,160x120 when the current sensor output is VGA.        |
| resolutionlist# |  | <width>x<height>,... | (optional) The available resolution of the current profile #. “#” is a number from 1 to the count of profiles.                   |
| vbitrates       |  | b1, b2, b3, ...      | available bitrate (kbps) list<br>for example, vbitrates=600,800,1000                                                             |
| goplengths      |  |                      | (optional) available GOP lengths list                                                                                            |
| qualitymodes    |  | CBR, Fixquality      | available quality mode list. for example, qualitymodes = CBR, Fixquality                                                         |
| framerates      |  |                      | available frame rates list                                                                                                       |
| frameratelist#  |  |                      | (optional) The available framerate of the current profile #. “#” is a number from 1 to the count of profiles.                    |
| qualities       |  |                      | available quality                                                                                                                |
| asamplerates    |  |                      | audio sample rates (kHz) list                                                                                                    |
| abitrates       |  |                      | audio bitrate (kbps) list                                                                                                        |
| micvol          |  |                      | available mic volume range from v1 to v2.<br>for example, micvol=0...100                                                         |
| cur_micvol      |  | An integer           | current mic volume. 0: also mean disabled.                                                                                       |
| speakervol      |  |                      | speaker volume range                                                                                                             |
| cur_speakervol  |  | An integer           | current speaker volume. 0: also mean disabled.                                                                                   |
| vprofileformat  |  | <Ver#>               | The current version is ‘1.5’:<br>This value indicates whether camera support /video/video.cgi or not. Please also refer to 4.1.7 |
| vprofilenum     |  | #                    | The total number of available video streams.                                                                                     |
| vprofile#       |  | <codec name>         | video profile # (# is a number from 1 to the count of profiles)                                                                  |
| vprofileurl#    |  |                      | the URL for video profile #                                                                                                      |
| vprofileres#    |  | <width>x<height>     | The resolution for video profile #. The resolution depends on the current sensor output.                                         |
| aprofilenum     |  | #                    | The total number of available audio streams.                                                                                     |
| aprofile#       |  | <codec name>         | audio profile # (# is a number from 1 to the count of profiles)                                                                  |
| aprofileurl#    |  |                      | the URL for audio profile #                                                                                                      |

### 3.3.2.get sensor output

request:

GET /config/sensor\_output.cgi



response:

| Name     | Value                                         | Description                                                                                                                                                                                                                             |
|----------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| supports | <sensor output name>,<sensor output name>,... | Available sensor output list. For example, supports=VGA,HDTV,4VGA                                                                                                                                                                       |
| current  | <sensor output name>                          | This value indicates the current sensor output. For example, current=4VGA. Changing the current sensor output result in the change of the resolutions and vprofileres# fields of stream_info.cgi and the resolution field of video.cgi. |

### 3.3.3.set sensor output

request:

GET/POST /config/sensor\_output.cgi

parameters:

current=<sensor output name>

response:

see the above table.

### 3.3.4.get video type

request:

GET/POST /config/video\_type.cgi

response:

| Name          | Value      | Description                                                                                                                                                                                |
|---------------|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| profilenumber | An integer | “1” meaning that single-stream<br>“2” meaning that dual-stream, and<br>“3” meaning that triple-stream.<br><br>If the profile has been used in the event(or recording), you can not set it. |
| aspectratio   | String     | 4:3 and 16:9                                                                                                                                                                               |

### 3.3.5.set video type

request:

GET/POST /config/video\_type.cgi

parameters:

see the above table.

response:

see the above table.

### 3.3.6.get video config

request:

GET /config/video.cgi

parameters:

profileid=<video profile number>

response:

| Name           | Value | Description                                                                                                                      |
|----------------|-------|----------------------------------------------------------------------------------------------------------------------------------|
| vprofileformat | Ver#  | The current version is ‘1.5’:<br>This value indicates whether camera support /video/video.cgi or not. Please also refer to 4.1.7 |
| profileid      | #     | profile number (# is a number from 1 to the count of profiles)                                                                   |

|             |                     |                                                                                                                                                                            |
|-------------|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| resolution  | <width>x<height>    | The resolution depends on the current sensor output.<br>Before you set the resolution of profile, you need to get the available video resolutions list by stream_info.cgi. |
| viewwindow  | <width>x<height>    | Digital PTZ resolution.                                                                                                                                                    |
| bitrate     | An integer          | in kbit/s                                                                                                                                                                  |
| codec       | MPEG4, MJPEG, H.264 | a video codec                                                                                                                                                              |
| goplength   | An integer          | the MPEG GOP length.                                                                                                                                                       |
| qualitymode | CBR, Fixquality     | quality mode                                                                                                                                                               |
| framerate   | 1 ... 30            | a frame rate in fps                                                                                                                                                        |
| quality     | # (0-100)           | available quality                                                                                                                                                          |

### 3.3.7.set video config

request:

GET/POST /config/video.cgi

parameters:

see the above table.

response:

see the above table.

### 3.3.8.sensors information

request:

GET /config/sensor\_info.cgi

response: (only supported parameters are displayed)

| Name            | Value                                                                        | Description                                                                                                     |
|-----------------|------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| brightness      | b1...b2 or b1,b2,b3                                                          | available brightness range or enumeration                                                                       |
| contrast        | c1...c2 or c1,c2,c3                                                          | available contrast range or enumeration                                                                         |
| saturation      | s1...s2 or s1,s2,s3                                                          | available saturation range or enumeration                                                                       |
| hue             | h1...h2 or h1,h2,h3                                                          | available hue range or enumeration                                                                              |
| whitebalance    | auto, fixed_indoor,<br>fixed_outdoor,<br>fixed_fluor, disabled               | available white balances list                                                                                   |
| maxexposuretime | m1...m2 or m1,m2,m3                                                          | a range of the maxexposuretime from 1/m <sub>1</sub> to 1/m <sub>2</sub> second or enumeration.                 |
| backlightcomp   | yes, no                                                                      | has backlight compensation                                                                                      |
| noisereduction  | off, low, high                                                               | a list of noise reduction level.                                                                                |
| autoexposure    | yes, no                                                                      | Indicate if camera support auto exposure function                                                               |
| autogainctrl    | yes, no                                                                      | Indicate if camera support auto gain control.                                                                   |
| inputsize       | <width>x<height>                                                             | Dimension of sensor size.                                                                                       |
| videooutformat  | auto: auto detect.<br>ntsc: NTSC<br>pal: PAL<br>pal-m: PAL M<br>pal-n: PAN-N | For the cameras which has an analog video output connector, this field indicates the format of the video signal |
| sharpness       | s1...s2 or s1,s2,s3                                                          | available sharpness range or enumeration.                                                                       |
| flicker         | auto, 50, 60                                                                 | anti flicker. auto or 50 or 60 Hz.                                                                              |
| mirror          | off, on                                                                      | disable/enable image flip horizontally                                                                          |
| flip            | off, on                                                                      | disable/enable image flip vertically                                                                            |
| color           | yes, no                                                                      | Select color mode or B/W mode                                                                                   |
| wds             | off, on                                                                      | disable/enable WDS                                                                                              |
| exposuremode    | Auto, Indoor, Outdoor, Night,<br>Moving, Low_noise, Customize1,              | available exposure mode list.                                                                                   |

|                |                                     |                                                                 |
|----------------|-------------------------------------|-----------------------------------------------------------------|
|                | Customize2, Customize3              |                                                                 |
| maxshutter     | s1...s2 or s1,s2,s3                 | a range of the shutter from 1/m1 to 1/m2 second or enumeration. |
| minshutter     | s1...s2 or s1,s2,s3                 | a range of the shutter from 1/m1 to 1/m2 second or enumeration. |
| maxgain        | g1...g2 or g1,g2,g3                 | available exposure mode gain range or enumeration.              |
| noisereduction | off, low, high, n1...n2 or n1,n2,n3 | available noise reduction range or enumeration.                 |
| wdrlevel       | w1...w2 or w1,w2,w3                 | available wdrlevel range or enumeration.                        |

### 3.3.9.get sensors config

Please call 3.3.6.sensor\_info.cgi to enumerate values of related parameters.

request:

GET /config/sensor.cgi

response: (only supported parameters are displayed.)

| Name            | Value                                                                               | Description                                                                                                                                                                                                                         |
|-----------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| brightness      | An integer                                                                          | the brightness                                                                                                                                                                                                                      |
| contrast        | An integer                                                                          | the contrast                                                                                                                                                                                                                        |
| saturation      | An integer                                                                          | the saturation                                                                                                                                                                                                                      |
| hue             | An integer                                                                          | the hue                                                                                                                                                                                                                             |
| whitebalance    | auto, fixed_indoor, fixed_outdoor, fixed_fluor, disabled                            | the white balance                                                                                                                                                                                                                   |
| flicker         | auto, 50, 60                                                                        | anti flicker. auto or 50 or 60 Hz.                                                                                                                                                                                                  |
| autoexposure    | yes, no                                                                             | enable or disable the auto exposure                                                                                                                                                                                                 |
| maxexposuretime | #                                                                                   | The divisor of maximum exposure time (1/# second). E.g. if the maximum exposure time is 1/10, then the value of this field is 10.                                                                                                   |
| backlightcomp   | yes, no                                                                             | backlight compensation. Will make darker objects in the foreground appear clearer if the background is very bright.                                                                                                                 |
| noisereduction  | off, low, high                                                                      | noise reduction level.                                                                                                                                                                                                              |
| mirror          | off, on                                                                             | disable/enable image flip horizontally                                                                                                                                                                                              |
| flip            | off, on                                                                             | disable/enable image flip vertically                                                                                                                                                                                                |
| autogainctrl    | yes, no                                                                             | enable or disable auto gain control function                                                                                                                                                                                        |
| color           | yes, no                                                                             | Select color mode or B/W mode                                                                                                                                                                                                       |
| videoinformat   | auto: auto detect.<br>ntsc: NTSC<br>pal: PAL<br>pal-m:PAL M<br>pal-n: PAN-N         | For video server, the input analog video signal could be one of many video formats such as NTSC or PAL. To let video server recognize the format of video input signal, the sensor module should be configured to match the format. |
| sharpness       | An integer                                                                          | the sharpness                                                                                                                                                                                                                       |
| wds             | off, on                                                                             | disable/enable WDS                                                                                                                                                                                                                  |
| exposuremode    | Auto, Indoor, Outdoor, Night, Moving, Low_noise, Customize1, Customize2, Customize3 | the exposure mode.                                                                                                                                                                                                                  |
| maxshutter      | #                                                                                   | the divisor of maximum shutter (1/# second). E.g. if the maximum shutter is 1/10, then the value of this field is 10.                                                                                                               |
| minshutter      | #                                                                                   | the divisor of minimum shutter (1/# second). E.g. if the minimum shutter is 1/10, then the value of this field is 10.                                                                                                               |
| maxgain         | An integer                                                                          | the max gain.                                                                                                                                                                                                                       |
| noisereduction  | off, low, high or integer                                                           | noise reduction level.                                                                                                                                                                                                              |
| wdrenable       | off, on                                                                             | Disable/enable WDR.                                                                                                                                                                                                                 |
| wdrlevel        | An integer                                                                          | The WDR level.                                                                                                                                                                                                                      |

### 3.3.10.set sensors config

request:

GET/POST /config/sensor.cgi

parameters:

see the above 2 tables to set the valid values.

response:  
see the above table.

### 3.3.11.get audio config

request:  
GET /config/audio.cgi

parameters:

profileid=<audio profile number>

response:

| Name       | Value                               | Description                                                          |
|------------|-------------------------------------|----------------------------------------------------------------------|
| profileid  | #                                   | audio profile number (# is a number from 1 to the count of profiles) |
| codec      | PCM, ADPCM, AMR, AAC, MU-LAW, A-LAW | the audio codec                                                      |
| samplerate | An integer                          | The clock rate for the audio sampling. (in kHz)                      |
| channel    | 1, 2                                | the audio channel number.                                            |
| bitrate    | An integer                          | The output bitrate. (in kbit/s)                                      |

### 3.3.12.set audio config

request:  
GET/POST /config/audio.cgi

parameters:

see the above table.

response:  
see the above table.

### 3.3.13.get microphone

request:  
GET /config/mic.cgi

response:

| Name   | Value   | Description                                  |
|--------|---------|----------------------------------------------|
| enable | no, yes | microphone disable/enable                    |
| volume |         | microphone volume. please refer 3.3.1 micvol |

### 3.3.14.set microphone

request:  
see the above table.

response:  
see the above table.

### 3.3.15.get speaker

request:  
GET /config/speaker.cgi

response:

| Name   | Value   | Description                                   |
|--------|---------|-----------------------------------------------|
| enable | no, yes | speaker disable/enable                        |
| volume |         | speaker volume. please refer 3.3.1 speakervol |

### 3.3.16.set speaker

request:

GET/POST /config/speaker.cgi

parameters:

see the above table.

response:

see the above table.

### 3.3.17.reset sensor to default configuration

request:

GET/POST /config/sensor\_reset.cgi

parameters:

reset=go

response:

| Name  | Value     | Description                |
|-------|-----------|----------------------------|
| reset | yes, fail | the result of sensor reset |

### 3.3.18.set IR LED

request:

GET/POST /config/irled.cgi

parameters:

| Name | Value      | Description           |
|------|------------|-----------------------|
| mode | on,<br>off | turn IR LED on or off |

response:

see the above table.

### 3.3.19.get ICR(Infrared Cut filter Removal) settings

request:

GET/POST /config/icr.cgi

response:

| Name      | Value                                          | Description                                                                                                                                      |
|-----------|------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| mode      | day,<br>night,<br>auto,<br>manual,<br>schedule | indicate whether icr is day, night, auto, manual or schedule                                                                                     |
| starttime | A time                                         | start time of schedule (in 24hr format "hh:mm", only when mode=schedule)<br>for example 07:30 means 7:30 am.<br>for example 19:30 means 7:30 pm. |
| endtime   | A time                                         | end time of schedule (in 24hr format "hh:mm", only when mode=schedule)<br>for example 07:30 means 7:30 am.<br>for example 19:30 means 7:30 pm.   |

### 3.3.20.set ICR(Infrared Cut filter Removal) settings

request:

see the above table.

response:

see the above table.

### 3.3.21.get stream authentication setting

request:

GET /config/stream\_auth.cgi

response:

| Name        | Value     | Description                                                        |
|-------------|-----------|--------------------------------------------------------------------|
| livevideo   | on<br>off | indicate whether it needs authentication to get live video stream. |
| snapshoturl | on<br>off | indicate whether it needs authentication to get a snapshot.        |

### 3.3.22.set stream authentication setting

request:

see the above table.

response:

see the above table.

Note: If the value of 'livevideo' is off, then the authentication for snapshot url will be turned off automatically.

### 3.3.23.query Privacy mask information

request:

GET /config/privacymask\_info.cgi

response:

| Name    | Value            | Description                              |
|---------|------------------|------------------------------------------|
| maxnum  | #                | maximum of privacy mask.                 |
| maxarea | <width>x<height> | privacy mask range of the maximum space. |

### 3.3.24.get Privacy mask

request:

GET /config/privacymask.cgi

response:

| Name    | Value    | Description                                                                                                                                              |
|---------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| enable# | no, yes  | disable/enable privacy mask window #.                                                                                                                    |
| area#   | A string | privacy mask window # in the format x,y,w,h<br>x,y is the coordinate. the 0,0 means the left top position.<br>w,h is the width and height of the window. |

### 3.3.25.set Privacy mask

request:

GET/POST /config/privacymask.cgi

parameters:

see the above table.

response:

see the above table.

## 3.4.network

### 3.4.1.get network config

request:

GET /config/network.cgi

response:

| Name             | Value         | Description                                                                         |
|------------------|---------------|-------------------------------------------------------------------------------------|
| dhcp             | off, on       | disable/Enable dynamic IP address assignment                                        |
| ip               | An IP address | IP address of static IP setting                                                     |
| netmask          | An IP address | subnet mask of static IP setting                                                    |
| gateway          | An IP address | default gateway of static IP setting                                                |
| dns1             | An IP address | primary DNS server of static IP setting                                             |
| dns2             | An IP address | secondary DNS server of static IP setting                                           |
| pppoe            | off, on       | use PPPoE                                                                           |
| pppoeuser        | A string      | PPPoE user name                                                                     |
| pppoepass        | A string      | PPPoE password                                                                      |
| ddns             | off, on       | disable/enable dynamic DNS service                                                  |
| ddnsprovider     |               | ID of the provider, see Table <a href="#">dynamic DNS service providers</a>         |
| ddnshost         | A string      | DDNS host name                                                                      |
| ddnsuser         | A string      | DDNS user name                                                                      |
| ddnspass         | A string      | DDNS password                                                                       |
| upnp             | off, on       | disable/enable UPnP                                                                 |
| httpport         | 1 ... 65535   | TCP port number for HTTP                                                            |
| httpexternalport | 1 ... 65535   | The external port number for UPnP NAT router to map the HTTP service port of camera |
| rtspport         | 1 ... 65535   | The port number of RTSP service                                                     |
| rtspexternalport | 1 ... 65535   | The external port number for UPnP NAT router to map the RTSP service port of camera |

### 3.4.2.set network config

request:

GET/POST /config/network.cgi

parameters:

see the above table.

response:

see the above table.

### 3.4.3.get PPPoE

request:

GET /config/pppoe.cgi

response:

| Name  | Value   | Description          |
|-------|---------|----------------------|
| pppoe | off, on | disable/enable PPPoE |
| user  |         | PPPoE user name      |

|      |                |
|------|----------------|
| pass | PPPoE password |
|------|----------------|

### 3.4.4.set PPPoE

request:

GET/POST /config/pppoe.cgi

parameters:

see the above table.

response:

see the above table.

### 3.4.5.get DDNS settings

request:

GET /config/ddns.cgi

response:

| Name     | Value   | Description                                                                 |
|----------|---------|-----------------------------------------------------------------------------|
| ddns     | off, on | disable/enable dynamic DNS service                                          |
| provider |         | ID of the provider, see Table <a href="#">dynamic DNS service providers</a> |
| host     |         | DDNS host name                                                              |
| user     |         | DDNS user name                                                              |
| pass     |         | DDNS password                                                               |

### 3.4.6.set DDNS

request:

GET/POST /config/ddns.cgi

parameters:

see the above table.

response:

see the above table.

### 3.4.7.get upnp information

request:

GET /config/upnp.cgi

response:

| Name   | Value   | Description                         |
|--------|---------|-------------------------------------|
| upnpav | off, on | disable/enable UPnP AV server.      |
| upnpcp | off, on | disable/enable UPnP CP port forward |

### 3.4.8.set upnp information

request:

GET /config/upnp.cgi

parameters:

see the above table.

response:

see the above table.

### 3.4.9.get TCP port number for HTTP

request:

GET /config/httpport.cgi



response:

| Name     | Value       | Description              |
|----------|-------------|--------------------------|
| httpport | 1 ... 65535 | TCP port number for HTTP |

### 3.4.10.set TCP port number for HTTP

request:

GET/POST /config/httpport.cgi

see the above table.

response:

see the above table.

### 3.4.11.get system wireless

request:

GET /config/wireless.cgi

response:

| Name       | Value                                 | Description                                                                                                                                                                                                                                                                                                                                                |
|------------|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| enable     | off, on                               | disable/enable wireless                                                                                                                                                                                                                                                                                                                                    |
| mode       | managed,<br>ad-hoc                    | The type of wireless network to associate with, managed (using an access point) or ad-hoc (not using an access point).                                                                                                                                                                                                                                     |
| essid      | A string                              | ESSID                                                                                                                                                                                                                                                                                                                                                      |
| chpatterns | A string                              | The pattern of available wireless channels. read-only.<br>1111000011110000 means channel 1,2,3,4,9,10,11,12 are available.                                                                                                                                                                                                                                 |
| channel    | 1 ... 16                              | wireless channel                                                                                                                                                                                                                                                                                                                                           |
| auth       | open,<br>shared,<br>WPA-PSK, WPA2-PSK | Authentication method. open system, shared key , WPA-PSK or WPA2-PSK                                                                                                                                                                                                                                                                                       |
| encryption | none,<br>WEP,<br>TKIP, AES            | when auth is open: none, WEP.<br>when auth is shared: WEP<br>when auth is WPA-PSK or WPA2-PSK: TKIP, AES                                                                                                                                                                                                                                                   |
| format     | hex,<br>ASCII                         | only used for WEP                                                                                                                                                                                                                                                                                                                                          |
| keylength  | 64,<br>128                            | WEP key length (bits)                                                                                                                                                                                                                                                                                                                                      |
| activekey  | 1 ... 4                               | Which WEP key to use when transmitting.                                                                                                                                                                                                                                                                                                                    |
| key1       |                                       | The keys must match the keys in the wireless access point. They could either be in hex format or in ASCII.<br>Hex: the string must be exactly 10 hex characters for 64-bit WEP and 26 hex characters for 128-bit WEP. (Hex chars are 0123456789ABCDEF)<br>ASCII: The string must be exactly 5 characters for 64-bit WEP and 13 characters for 128-bit WEP. |
| key2       |                                       |                                                                                                                                                                                                                                                                                                                                                            |
| key3       |                                       |                                                                                                                                                                                                                                                                                                                                                            |
| key4       |                                       |                                                                                                                                                                                                                                                                                                                                                            |
| passphrase |                                       | WPA passphrase                                                                                                                                                                                                                                                                                                                                             |

### 3.4.12.set system wireless

request:

GET/POST /config/wireless.cgi

parameters:

see the above table.

response:

see the above table.

### 3.4.13.get current wireless connection condition

request:

GET /config/wlansignal.cgi

response:

| Name   | Value   | Description                              |
|--------|---------|------------------------------------------|
| signal | 0...100 | Current wireless channel signal strength |

### 3.4.14.do wireless site survey

request:

GET /config/wlansurvey.cgi

response: (1 site)

| Name       | Value                                 | Description                                                                                              |
|------------|---------------------------------------|----------------------------------------------------------------------------------------------------------|
| ssid       | A string                              | SSID                                                                                                     |
| signal     | 0...100                               | The signal strength indicator of wireless AP.                                                            |
| mode       | Ad-hoc<br>infrastructure              | Wireless mode                                                                                            |
| channel    | 1 ... 16                              | wireless channel                                                                                         |
| auth       | open,<br>shared,<br>WPA-PSK, WPA2-PSK | Authentication method. open system, shared key , WPA-PSK or WPA2-PSK                                     |
| encryption | none,<br>WEP,<br>TKIP, AES            | when auth is open: none, WEP.<br>when auth is shared: WEP<br>when auth is WPA-PSK or WPA2-PSK: TKIP, AES |

note:

Each wireless AP (access point) found has several attributes such as the above table. 'ssid' is the first attribute of any one wireless AP. The camera output these attributes of all found wireless AP in sequence.

### 3.4.15.get HTTPS configuration

request:

GET/POST /config/https.cgi

response:

| Name                          | Value       | Description                                                                                  |
|-------------------------------|-------------|----------------------------------------------------------------------------------------------|
| https_enable                  | no, yes     | HTTPS function is disabled or enabled                                                        |
| https_only                    | no, yes     | HTTP function is disabled or enabled(Indicate whether only HTTPS function is enabled or not) |
| certificate_country           | A string    | Country name for self-signed certificate(2 letter code)                                      |
| certificate_state             | A string    | State or province name for self-signed certificate                                           |
| certificate_locality          | A string    | Locality name for self-signed certificate                                                    |
| certificate_organization      | A string    | Organization name for self-signed certificate                                                |
| certificate_organization_unit | A string    | Organizational unit name for self-signed certificate                                         |
| certificate_common_name       | A string    | Common name for self-signed certificate                                                      |
| certificate_validity          | 1 ... 65535 | Number of days(validity) for self-signed certificate                                         |

### 3.4.16.set HTTPS configuration

request:

GET/POST /config/https.cgi

parameters:

see the above table.

response:

see the above table.

### 3.4.17.list all the IP access list

request:

GET /config/acces\_list.cgi

response:

| Name      | Value                             | Description               |
|-----------|-----------------------------------|---------------------------|
| maxallow  | An integer                        | Maximum number of access. |
| allowlist | <An IP address~An IP address>,... | All allow IP range.       |
| maxdeny   | An integer                        | Maximum number of access. |
| denylist  | <An IP address~An IP address>,... | All deny IP range.        |

### 3.4.18.add, delete acces IP

request:

GET /config/acces.cgi

parameters:

| Name  | Value                         | Description                                                                                                                                                           |
|-------|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| type  | allow, deny                   | Select the type of setting.                                                                                                                                           |
| act   | add, del                      | Select the action to perform content.                                                                                                                                 |
| range | An IP address ~ An IP address | This parameter applies only in the "act=add".                                                                                                                         |
| index | An integer                    | This parameter applies only in the "act=del". According to this index value in the IP access list to delete the corresponding position. The minimum index value is 0. |

response:

See the above table.

Set parameters (range) successful will return the IP address of the original set value, if it fails, return "empty string"(e.g. range=\r\n).Set parameters (index) successful will return index value, if it fails, return "-1"(e.g. index=-1\r\n).

## 3.5.Motion Detection

### 3.5.1.get motion detection

There are 2 possible types of motion detection dependent on your IP camera hardware.

request:

GET /config/motion.cgi

response:

macro block type:

| Name        | Value        | Description                                                                                                                                                                                                                                                                                                                                                                     |
|-------------|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| enable      | no, yes      | disable/enable motion detection                                                                                                                                                                                                                                                                                                                                                 |
| mbmask      | A hex string | The macro block mask hex string of the native screen resolution which is calculated linearly from left to right then top to bottom. The size of one macro block depends on the video resolution. However, no matter the resolution of video is, the number of macro block is always 40x30. That is there is 40 block in extension of the width of video and 30 block in height. |
| sensitivity | 0 ... 100    | sensitivity                                                                                                                                                                                                                                                                                                                                                                     |
| percentage  | 0...100      | percentage                                                                                                                                                                                                                                                                                                                                                                      |
| pir         | no, yes      | disable/enable PIR                                                                                                                                                                                                                                                                                                                                                              |

window type:

| Name | Value | Description |
|------|-------|-------------|
|------|-------|-------------|

|             |           |                                                                                                                                                              |
|-------------|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| totalnum    | #         | total motion detection window numbers. read-only.                                                                                                            |
| sensitivity | 0 ... 100 | sensitivity                                                                                                                                                  |
| percentage  | 0...100   | percentage                                                                                                                                                   |
| enable#     | no, yes   | disable/enable motion detection window #                                                                                                                     |
| mdw#        | A string  | motion detection window # in the format x,y,w,h<br>x,y is the coordinate. the 0,0 means the left top position.<br>w,h is the width and height of the window. |
| pir         | no, yes   | disable/enable PIR                                                                                                                                           |

### 3.5.2.set motion detection

request:

GET/POST /config/motion.cgi

parameters:

see the above table.

response:

see the above table.

### 3.5.3.get Recorder action

request:

GET /cgi/admin/recorder.cgi

|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Response:<br>Represented by XML | <pre> &lt;config&gt;   &lt;record&gt;     &lt;enable&gt;0&lt;/enable&gt;     &lt;profileID&gt;0&lt;/profileID&gt;     &lt;continuous&gt;0&lt;/continuous&gt;     &lt;prerecord&gt;0&lt;/prerecord&gt;     &lt;postrecord&gt;0&lt;/postrecord&gt;     &lt;keepSpace&gt;100&lt;/keepSpace&gt;     &lt;fileLenMin&gt;1&lt;/fileLenMin&gt;     &lt;recycle&gt;0&lt;/recycle&gt;     &lt;recordTo&gt;       &lt;toSamba&gt;         &lt;enable&gt;0&lt;/enable&gt;         &lt;anonymous&gt;1&lt;/anonymous&gt;         &lt;user&gt;&lt;/user&gt;         &lt;password&gt;&lt;/password&gt;         &lt;server&gt;&lt;/server&gt;         &lt;shareFolder&gt;&lt;/shareFolder&gt;       &lt;/toSamba&gt;       &lt;toUSB&gt;         &lt;enable&gt;1&lt;/enable&gt;       &lt;/toUSB&gt;     &lt;/recordTo&gt;   &lt;/record&gt;   &lt;schedule&gt;     &lt;enable&gt;0&lt;/enable&gt;     &lt;profileName&gt;Record&lt;/profileName&gt;     &lt;item01&gt;0,0,0,1,0,0&lt;/item01&gt;     &lt;item02&gt;1,0,0,2,0,0&lt;/item02&gt;     &lt;item03&gt;2,0,0,3,0,0&lt;/item03&gt;     &lt;item04&gt;3,0,0,4,0,0&lt;/item04&gt;     &lt;item05&gt;4,0,0,5,0,0&lt;/item05&gt;     &lt;item06&gt;5,0,0,6,0,0&lt;/item06&gt;     &lt;item07&gt;6,0,0,0,0,0&lt;/item07&gt;     &lt;itemSize&gt;7&lt;/itemSize&gt;   &lt;/schedule&gt;   &lt;triggerBy&gt;     &lt;byMotion&gt;0&lt;/byMotion&gt;     &lt;byIn1&gt;0&lt;/byIn1&gt;   &lt;/triggerBy&gt; &lt;/config&gt; </pre> |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|  |                                                                                              |
|--|----------------------------------------------------------------------------------------------|
|  | <pre> &lt;byln2&gt;0&lt;/byln2&gt; &lt;/triggerBy&gt; &lt;/record&gt; &lt;/config&gt; </pre> |
|--|----------------------------------------------------------------------------------------------|

### 3.5.4.set Recorder action

request:

GET/POST /cgi/admin/recorder.cgi

parameters:

| Name         | Value                                                                       | Description                                                                                                                                                                                                                                                                                                                                 |
|--------------|-----------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| recordEnable | 0 or 1                                                                      | 0:disable 1:Enable recording action                                                                                                                                                                                                                                                                                                         |
| profileID    | #                                                                           | Id of recording profile.                                                                                                                                                                                                                                                                                                                    |
| prerecord    | #                                                                           | pre-recording time in second (0~15)                                                                                                                                                                                                                                                                                                         |
| postrecord   | #                                                                           | post-recording time in second (0~15)                                                                                                                                                                                                                                                                                                        |
| keepSpace    | #                                                                           | keep 32~999999 Mbyte for hard disk or SD card                                                                                                                                                                                                                                                                                               |
| fileLenMin   | 1 ... 60                                                                    | Separate the recording file in 1 ... 60 minute(s). For the limitation of the maximum value of each model, please reference by each model support table.                                                                                                                                                                                     |
| recycle      | 0 or 1                                                                      | 0: disable 1: enable, recording recycle while the CIFS disk or USB is full                                                                                                                                                                                                                                                                  |
| toUSB        | 0 or 1                                                                      | 0: disable 1: enable, record to USB or Samba. These two parameters must be both given and enable only one of it.                                                                                                                                                                                                                            |
| toSamba      |                                                                             |                                                                                                                                                                                                                                                                                                                                             |
| byMotion     | 0 or 1                                                                      | 0: disable 1: enable, events to trigger recording. These parameters must be all given and could be multiple enabled.                                                                                                                                                                                                                        |
| byIn1        |                                                                             |                                                                                                                                                                                                                                                                                                                                             |
| byIn2        |                                                                             |                                                                                                                                                                                                                                                                                                                                             |
| continuous   | 0 or 1                                                                      | 0: disable 1: enable, record continuous or by schedule. These two parameters must be both given, and enable only one of it.                                                                                                                                                                                                                 |
| schedule     |                                                                             |                                                                                                                                                                                                                                                                                                                                             |
| anonymous    | 0 or 1                                                                      | 0: login as account 1: login as anonymous                                                                                                                                                                                                                                                                                                   |
| user         | A string                                                                    | at most 30 characters, can accept => !#\$%()*+,-./:;=?@[\]^_`{ }~                                                                                                                                                                                                                                                                           |
| password     | A string                                                                    | at most 30 characters, can accept => !#\$%()*+,-./:;=?@[\]^_`{ }~                                                                                                                                                                                                                                                                           |
| server       | A string                                                                    | the server ip, at most 30 characters, can accept => _-.                                                                                                                                                                                                                                                                                     |
| shareFolder  | A string                                                                    | at most 30 characters, can accept => _-                                                                                                                                                                                                                                                                                                     |
| item01       | A string<br>[A:0...6],[B:0...23],[C:0...59],[D:0...6],[E:0...23],[F:0...59] | A: start day(0:Sunday ... 6:Saturday) B: start hour C: start minute D: end day(0:Sunday ... 6:Saturday) E: stop hour F: stop minute<br>ex. 1,1,1,1,2,2 means schedule record from Monday AM:01:01 to Monday AM:02:02<br>ex. 1,0,0,2,0,0 mean schedule record for a whole day of Monday.<br>ps. This item should be set relative to itemSize |
| item02       |                                                                             |                                                                                                                                                                                                                                                                                                                                             |
| item03       |                                                                             |                                                                                                                                                                                                                                                                                                                                             |
| item04       |                                                                             |                                                                                                                                                                                                                                                                                                                                             |
| item05       |                                                                             |                                                                                                                                                                                                                                                                                                                                             |
| item06       |                                                                             |                                                                                                                                                                                                                                                                                                                                             |
| item07       |                                                                             |                                                                                                                                                                                                                                                                                                                                             |
| itemSize     | #                                                                           | save the number of schedule items(0~7), ex itemSize = 3, item01~item03's value will be saved.                                                                                                                                                                                                                                               |

response:

see the 3.5.3 table. An extra result tag will also be presented:

|                               |                                                                                                                |
|-------------------------------|----------------------------------------------------------------------------------------------------------------|
| Result:<br>Represented by XML | <pre> &lt;config&gt;   &lt;result&gt;     &lt;code&gt;ok&lt;/code&gt;   &lt;/result&gt; &lt;/config&gt; </pre> |
|-------------------------------|----------------------------------------------------------------------------------------------------------------|

|             |                     |                                                  |
|-------------|---------------------|--------------------------------------------------|
| Result code | 1. ok               | Success                                          |
|             | 2. invalidParameter | Fail in cgi checking, please check setting rules |
|             | 3. updateFailed     | Fail in cgi setting, please check setting rules  |

### 3.5.5.get Snapshot action

request:

GET /cgi/admin/adv\_snapshot\_cont.cgi

|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Response:<br>Represented by XML | <pre> &lt;config&gt;   &lt;mail&gt;     &lt;smtpEnable1&gt;0&lt;/smtpEnable1&gt;     &lt;smtpServer1&gt;&lt;/smtpServer1&gt;     &lt;smtpPort1&gt;25&lt;/smtpPort1&gt;     &lt;smtpUser1&gt;&lt;/smtpUser1&gt;     &lt;smtpPass1&gt;&lt;/smtpPass1&gt;     &lt;receiver1&gt;&lt;/receiver1&gt;     &lt;sender1&gt;&lt;/sender1&gt;   &lt;/mail&gt;   &lt;ftp&gt;     &lt;ftpEnable1&gt;0&lt;/ftpEnable1&gt;     &lt;ftpServer1&gt;&lt;/ftpServer1&gt;     &lt;ftpPort1&gt;21&lt;/ftpPort1&gt;     &lt;ftpUser1&gt;&lt;/ftpUser1&gt;     &lt;ftpPass1&gt;&lt;/ftpPass1&gt;     &lt;folder1&gt;&lt;/folder1&gt;     &lt;ftpInterval&gt;1&lt;/ftpInterval&gt;     &lt;passive1&gt;1&lt;/passive1&gt;     &lt;fixFile1&gt;DCS-2102&lt;/fixFile1&gt;   &lt;/ftp&gt;   &lt;snapshot&gt;     &lt;enable&gt;0&lt;/enable&gt;     &lt;continuous&gt;0&lt;/continuous&gt;     &lt;schedule&gt;       &lt;enable&gt;0&lt;/enable&gt;       &lt;profileName&gt;Snapshot&lt;/profileName&gt;       &lt;item01&gt;0,0,0,1,0,0&lt;/item01&gt;       &lt;item02&gt;1,0,0,2,0,0&lt;/item02&gt;       &lt;item03&gt;2,0,0,3,0,0&lt;/item03&gt;       &lt;item04&gt;3,0,0,4,0,0&lt;/item04&gt;       &lt;item05&gt;4,0,0,5,0,0&lt;/item05&gt;       &lt;item06&gt;5,0,0,6,0,0&lt;/item06&gt;       &lt;item07&gt;6,0,0,0,0,0&lt;/item07&gt;       &lt;itemSize&gt;7&lt;/itemSize&gt;     &lt;/schedule&gt;     &lt;triggerBy&gt;       &lt;byMotion&gt;0&lt;/byMotion&gt;       &lt;byIn1&gt;0&lt;/byIn1&gt;       &lt;byIn2&gt;0&lt;/byIn2&gt;     &lt;/triggerBy&gt;     &lt;snapTo&gt;       &lt;toFtp&gt;1&lt;/toFtp&gt;       &lt;toSmtt&gt;1&lt;/toSmtt&gt;     &lt;/snapTo&gt;   &lt;/snapshot&gt; &lt;/config&gt; </pre> |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### 3.5.6.set Snapshot action

request:

GET/POST /cgi/admin/adv\_snapshot\_cont.cgi

parameters:

| Name | Value | Description |
|------|-------|-------------|
|------|-------|-------------|

|             |                                                                             |                                                                                                                                                                                                                                                                                                                                                    |
|-------------|-----------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| enable      | 0 or 1                                                                      | 0:disable 1:Enable snapshot function                                                                                                                                                                                                                                                                                                               |
| toFTP       | 0 or 1                                                                      | 0: disable 1: enable. Snapshot to FTP or SMTP. These two parameters must be both given and could be multiple enabled.                                                                                                                                                                                                                              |
| toSmtp      |                                                                             |                                                                                                                                                                                                                                                                                                                                                    |
| continuous  | 0 or 1                                                                      | 0: disable 1: enable, These two parameters must be both given, and enable only one of it. Please note only FTP support continuous snapshot.                                                                                                                                                                                                        |
| schedule    |                                                                             |                                                                                                                                                                                                                                                                                                                                                    |
| byMotion    | 0 or 1                                                                      | 0: disable 1: enable, events (multiselection) to trigger snapshot. These parameters must be all given.                                                                                                                                                                                                                                             |
| byIn1       |                                                                             |                                                                                                                                                                                                                                                                                                                                                    |
| byIn2       |                                                                             |                                                                                                                                                                                                                                                                                                                                                    |
| smtpEnable1 | 0 or 1                                                                      | 0:disable 1:Enable snapshot to SMTP1                                                                                                                                                                                                                                                                                                               |
| smtpServer1 | A string                                                                    | SMTP Server host name or IP address, at most 60 characters, can accept => _ - .                                                                                                                                                                                                                                                                    |
| smtpUser1   | A string                                                                    | SMTP user name, at most 30 characters, can accept<br>=> !"#%&'()*+,-./:;<=>?@[\\]^_`{ }~                                                                                                                                                                                                                                                           |
| smtpPass1   | A string                                                                    | SMTP password, at most 30 characters, can accept<br>=> !"#%&'()*+,-./:;<=>?@[\\]^_`{ }~                                                                                                                                                                                                                                                            |
| receiver1   | A string                                                                    | to which email address, at most 60 characters, can accept<br>=> !"#%&'()*+,-./:;<=>?@[\\]^_`{ }~                                                                                                                                                                                                                                                   |
| sender1     | A string                                                                    | from which email address, at most 60 characters, can accept<br>=> !"#%&'()*+,-./:;<=>?@[\\]^_`{ }~                                                                                                                                                                                                                                                 |
| ftpEnable1  | 0 or 1                                                                      | 0:disable 1:Enable snapshot to FTP1                                                                                                                                                                                                                                                                                                                |
| ftpServer1  | A string                                                                    | FTP server host name or IP address, at most 60 characters, can accept<br>=> _ - .                                                                                                                                                                                                                                                                  |
| ftpPort1    | #                                                                           | FTP connect server port(1~65535)                                                                                                                                                                                                                                                                                                                   |
| ftpUser1    | A string                                                                    | FTP login user name, at most 30 characters, can accept<br>=> !"#%&'()*+,-./:;<=>?@[\\]^_`{ }~                                                                                                                                                                                                                                                      |
| ftpPass1    | A string                                                                    | FTP password, at most 30 characters, can accept<br>=> !"#%&'()*+,-./:;<=>?@[\\]^_`{ }~                                                                                                                                                                                                                                                             |
| folder1     | A string                                                                    | FTP init path, at most 30 characters, can not accept =><br>~!@#%^&'()*+{}`=[]; ', / \   *                                                                                                                                                                                                                                                          |
| ftpInterval | #                                                                           | the time interval for continue snapshot to FTP in second.(1~86400)                                                                                                                                                                                                                                                                                 |
| passive1    | 0 or 1                                                                      | 0:disable 1:Enable FTP passive mode                                                                                                                                                                                                                                                                                                                |
| fixFile1    | A string                                                                    | the prefix name of picture, at most 30 characters, can not accept =><br>~!@#%^&'()*+{}`=[]; ', / \   *                                                                                                                                                                                                                                             |
| item01      | A string<br>[A:0...6],[B:0...23],[C:0...59],[D:0...6],[E:0...23],[F:0...59] | A: start day(0:Sunday ... 6:Saturday) B: start hour C: start minute<br>D: end day(0:Sunday ... 6:Saturday) E: stop hour F: stop minute<br>ex. 1,1,1,1,2,2 means schedule snapshot from Monday AM:01:01 to Monday AM:02:02<br>ex. 1,0,0,2,0,0 mean schedule snapshot for a whole day of Monday.<br>ps. This item should be set relative to itemSize |
| item02      |                                                                             |                                                                                                                                                                                                                                                                                                                                                    |
| item03      |                                                                             |                                                                                                                                                                                                                                                                                                                                                    |
| item04      |                                                                             |                                                                                                                                                                                                                                                                                                                                                    |
| item05      |                                                                             |                                                                                                                                                                                                                                                                                                                                                    |
| item06      |                                                                             |                                                                                                                                                                                                                                                                                                                                                    |
| item07      |                                                                             |                                                                                                                                                                                                                                                                                                                                                    |
| itemSize    | #                                                                           | save the number of schedule items(0~7), ex itemSize = 3, item01~item03's value will be saved.                                                                                                                                                                                                                                                      |

response:

see the 3.5.5 table. An extra result tag will also be presented:

|                               |                                                                                                              |                                                  |
|-------------------------------|--------------------------------------------------------------------------------------------------------------|--------------------------------------------------|
| Result:<br>Represented by XML | <pre>&lt;config&gt;   &lt;result&gt;     &lt;code&gt;ok&lt;/code&gt;   &lt;/result&gt; &lt;/config&gt;</pre> |                                                  |
| Result code                   | 1. ok                                                                                                        | Success                                          |
|                               | 2. invalidParameter                                                                                          | Fail in cgi checking, please check setting rules |
|                               | 3. updateFailed                                                                                              | Fail in cgi setting, please check setting rules  |

### 3.5.7.get Alarm out action

request:  
GET /cgi/admin/adv\_do.cgi

|                                 |                                                                                                                                                                                                                                                                                                                                              |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Response:<br>Represented by XML | <pre>&lt;config&gt;   &lt;digitalOutput&gt;     &lt;toOut1&gt;       &lt;enable&gt;0&lt;/enable&gt;       &lt;triggerBy&gt;         &lt;byMotion&gt;0&lt;/byMotion&gt;         &lt;byIn1&gt;0&lt;/byIn1&gt;         &lt;byIn2&gt;0&lt;/byIn2&gt;       &lt;/triggerBy&gt;     &lt;/toOut1&gt;   &lt;/digitalOutput&gt; &lt;/config&gt;</pre> |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### 3.5.8.set alarm out action

request:  
GET/POST /cgi/admin/adv\_do.cgi

parameters:

| Name         | Value  | Description                                                         |
|--------------|--------|---------------------------------------------------------------------|
| toOut1       | 0 or 1 | alarm to output 1                                                   |
| out1ByMotion | 0 or 1 | 0: disable 1: enable, events (multiselection) to trigger alarm out1 |
| out1ByIn1    |        |                                                                     |
| out1ByIn2    |        |                                                                     |

response:  
see the 3.5.7 table. An extra result tag will also be presented:

|                               |                                                                                                              |                                                  |
|-------------------------------|--------------------------------------------------------------------------------------------------------------|--------------------------------------------------|
| Result:<br>Represented by XML | <pre>&lt;config&gt;   &lt;result&gt;     &lt;code&gt;ok&lt;/code&gt;   &lt;/result&gt; &lt;/config&gt;</pre> |                                                  |
| Result code                   | 1. ok                                                                                                        | Success                                          |
|                               | 2. invalidParameter                                                                                          | Fail in cgi checking, please check setting rules |
|                               | 3. updateFailed                                                                                              | Fail in cgi setting, please check setting rules  |



## 3.6.system tools

### 3.6.1.get digital input/output

This CGI is an one-shot command, which only return current status of IP camera. If you need to minitor camera status for a long time, please use 6.1.2. notify\_stream.cgi instead.

request:

GET /config/io.cgi

response: (only supported inputs and outputs are displayed)

| Name | Value   | Description          |
|------|---------|----------------------|
| in1  | off, on | Digital input set 1  |
| in2  | off, on | Digital input set 2  |
| in3  | off, on | Digital input set 3  |
| in4  | off, on | Digital input set 4  |
| out1 | off, on | Digital output set 1 |
| out2 | off, on | Digital output set 2 |
| out3 | off, on | Digital output set 3 |
| out4 | off, on | Digital output set 4 |

### 3.6.2.set digital output

You can only set the available digital outputs, inputs are read-only.

request:

GET/POST /config/io.cgi

parameters:

|      |         |                      |
|------|---------|----------------------|
| out1 | off, on | Digital output set 1 |
| out2 | off, on | Digital output set 2 |
| out3 | off, on | Digital output set 3 |
| out4 | off, on | Digital output set 4 |

response:

see the above table.

### 3.6.3.get LED

request:

GET /config/led.cgi

response:

| Name | Value   | Description                                |
|------|---------|--------------------------------------------|
| led  | on, off | enable or disable the special purpose LED. |

### 3.6.4.set LED

request:

GET/POST /config/led.cgi

parameters:

see the above table.

response:

see the above table.

### 3.6.5.firmware upgrade

request:

POST /config/firmwareupgrade.cgi

The file content is provided in the HTTP body according to the format given in RFC 1867. The body is created automatically by the browser if using HTML form with input type "file".

Example:

```
POST /config/firmwareupgrade.cgi HTTP/1.0\r\n
Content-Type: multipart/form-data; boundary=AsCg5y\r\n
Content-Length: <content length>\r\n
\r\n
--AsCg5y\r\n
Content-Disposition: form-data; name="update.bin"; filename="update.bin"\r\n
Content-Type: application/octet-stream\r\n
\r\n
<firmware file content>
\r\n
--AsCg5y--\r\n
```

response:

| Name    | Value    | Description                        |
|---------|----------|------------------------------------|
| upgrade | ok, fail | the upgrade was successful or fail |

Note:

You can use web browser (e.g. Microsoft Internet Explorer or FireFox) to transport firmware to IP camera. To do this, you should write a HTML file with a form architecture to post firmware file to camera. For example:

```
<!-- saved from url=(0022)http://internet.e-mail -->
<html>

<head>
<script language="JavaScript" type="text/javascript">
function sendUpdate()
{
    var updateForm = document.updateForm;
    document.updateForm.action = "http://" + camip.value + "/config/firmwareupgrade.cgi";
    updateForm.submit();
}
</script>
</head>

<body>
Input camera ip (ex. 192.168.1.1): <input name="camip" type="text" id="camip" value=""/>
<form enctype="multipart/form-data" method="post" action="" name="updateForm">
    Choose firmware file: <input name="upload" type="file" id="upload" value=""/>
    and click
    <input name="submit6" value="commit" type="button" onclick="sendUpdate()" />
</form>
</body>

</html>
```

### 3.6.6.reboot the camera

request:

GET/POST /config/system\_reboot.cgi

parameters:

reboot=go

response:

| Name   | Value     | Description                       |
|--------|-----------|-----------------------------------|
| reboot | yes, fail | the reboot was successful or fail |

### 3.6.7.reset all configurations to the factory default

request:

GET/POST /config/system\_reset.cgi

parameters:

reset=go

response:

| Name  | Value     | Description                      |
|-------|-----------|----------------------------------|
| reset | yes, fail | the reset was successful or fail |

### 3.6.8.get RS-485 settings

request:

GET /config/rs485.cgi

response:

| Name        | Value                                               | description                                                                                                                                                         |
|-------------|-----------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| enable      | no, yes                                             | disable/enable RS-485                                                                                                                                               |
| proto       | custom, Dyna, Lilin, Lilin2, PelcoD, PelcoP         | protocol type                                                                                                                                                       |
| devid       |                                                     | device ID of the RS-485 slave device.<br>Dyna: 1 ... 223<br>Lilin: 1 ... 64<br>Lilin2: 0 ... 255<br>PelcoD: 1 ... 255<br>PelcoP: 1 ... 32<br>custom: not applicable |
| baudrate    | 1200, 2400, 4800, 9600, 19200, 38400, 57600, 115200 | custom baud rate                                                                                                                                                    |
| databits    | 7, 8                                                | custom data bits                                                                                                                                                    |
| parity      | None, Even, Odd                                     | custom parity                                                                                                                                                       |
| stopbits    | 1, 2                                                | custom stop bits                                                                                                                                                    |
| home        |                                                     | custom home command                                                                                                                                                 |
| up          |                                                     | custom up command                                                                                                                                                   |
| down        |                                                     | custom down command                                                                                                                                                 |
| left        |                                                     | custom left command                                                                                                                                                 |
| right       |                                                     | custom right command                                                                                                                                                |
| stop        |                                                     | custom stop command                                                                                                                                                 |
| stoppattern | A string                                            | whether to use the custom stop command for custom command 1, 2, 3, 4.<br>0101 means custom command 2 and 4 with stop command.                                       |
| cmdname1    |                                                     | custom command1 name                                                                                                                                                |
| cmdname2    |                                                     | custom command2 name                                                                                                                                                |
| cmdname3    |                                                     | custom command3 name                                                                                                                                                |
| cmdname4    |                                                     | custom command4 name                                                                                                                                                |
| cmdstr1     |                                                     | custom command1 string                                                                                                                                              |
| cmdstr2     |                                                     | custom command2 string                                                                                                                                              |
| cmdstr3     |                                                     | custom command3 string                                                                                                                                              |
| cmdstr4     |                                                     | custom command4 string                                                                                                                                              |
| delaytime   |                                                     | the interval of time between two consecutive command string being executed(in millisecond, for example 300ms).                                                      |

### 3.6.9.set RS-485 settings

request:

GET/POST /config/rs485.cgi

see the above table.

response:

see the above table.

### 3.6.10.get Privacy mode settings

request:

GET /config/privacy.cgi

response:

| Name   | Value   | Description                          |
|--------|---------|--------------------------------------|
| enable | no, yes | disable/enable Privacy mode          |
| manual | on, off | Turn on or turn off the privacy mode |

### 3.6.11.set Privacy mode settings

request:

GET/POST /config/ privacy.cgi

see the above table.

response:

see the above table.

### 3.6.12.get TV output

request:

GET /config/tvoutput.cgi

response:

| Name | Value   | Description                                                      |
|------|---------|------------------------------------------------------------------|
| out  | on, off | Disable/enable this cable connection into the TV output adapter. |

### 3.6.13.set TV output

request:

GET/POST /config/tvoutput.cgi

see the above table.

response:

see the above table.

### 3.6.14.get DC power

request:

GET /config/dcpower.cgi

response:

| Name      | Value                   | Description                                                                                                                                      |
|-----------|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| mode      | on, off, auto, schedule | Disable/enable the camera's DC power port.                                                                                                       |
| starttime | A time                  | start time of schedule (in 24hr format "hh:mm", only when mode=schedule)<br>for example 07:30 means 7:30 am.<br>for example 19:30 means 7:30 pm. |

|         |        |                                                                                                                                                  |
|---------|--------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| endtime | A time | start time of schedule (in 24hr format "hh:mm", only when mode=schedule)<br>for example 07:30 means 7:30 am.<br>for example 19:30 means 7:30 pm. |
|---------|--------|--------------------------------------------------------------------------------------------------------------------------------------------------|

### 3.6.15.set DC power

request:

GET/POST /config/dcpower.cgi

see the above table.

response:

see the above table.

### 3.6.16.get device timestamp

request:

GET /config/timestamp.cgi

response:

| Name     | Value    | Description                                     |
|----------|----------|-------------------------------------------------|
| enable   | on, off  | Disable/enable the camera's time stamp (label). |
| showtime | on, off  | Disable/enable the camera's time stamp (time).  |
| label    | A String | The time stamp on the image.                    |

### 3.6.17.set device timestamp

request:

GET/POST /config/timestamp.cgi

see the above table.

response:

see the above table.

## 3.7.SD card operation

### 3.7.1.get information of SD card

This is an one-shot request, client of NIPCA could use this request to get a summary of SD card inserted in the camera. For getting up-to-date status of SD card, client should request to notify\_stream.cgi (see 6.1.2), instead of pooling this cgi frequently.

request:

GET /config/sdcard.cgi

response: (only supported inputs and outputs are displayed)

| Name    | Value                                                                                    | Description                            |
|---------|------------------------------------------------------------------------------------------|----------------------------------------|
| status  | ready, protected, full, invalid, over_capacity, need_reinitialize, formatting, recording | Current status of SD Card              |
| total   | integer                                                                                  | Total size of SD Card in KBytes        |
| used    | integer                                                                                  | Used size of SD Card in KBytes         |
| free    | integer                                                                                  | Free size of SD Card in KBytes         |
| picture | integer                                                                                  | How many sub-folders in picture folder |
| video   | integer                                                                                  | How many sub-folders in video folder   |

### 3.7.2.format SD card

request:

GET /config/sdcard\_format.cgi

parameters:

| Name   | Value | Description                   |
|--------|-------|-------------------------------|
| format | go    | Do format or query SD status. |

response:

see the 3.7.1 table.

### 3.7.3.list items of SD card

request:

GET /config/sdcard\_list.cgi

parameters:

| Name     | Value          | Description                                                                                         |
|----------|----------------|-----------------------------------------------------------------------------------------------------|
| type     | picture, video | Which type of files would like to see                                                               |
| path     | string         | Indicate which files of path you want to get.                                                       |
| page     | integer        | This command would list files of a page. You can indicate which list of page you would like to see. |
| pagesize | integer        | How many files in a page. Value=5, 10, 15, 20, 50 or 100.                                           |

response:

| Name       | Value                                    | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|------------|------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| sd_status  | ready,<br>invalid,<br>protected,<br>full | Status of SD Card.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| type       | picture, video                           | Which type of files would like to see                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| path       | string                                   | Indicate current path to be examined                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| page       | integer                                  | This command would list files of a page. You can indicate which list of page you would like to see.<br>For example:<br>If I have 40 items (file or folder) in `path` folder and would like to list the files in page 1 where I assume the number of items in each page is 20.<br>The request url may be:<br>GET<br>/config/sdcard_list.cgi?type=picture&path=/20080229/00&page=1&pagesize=20<br>Where:<br>type=picture: list the items in path 'picture'<br>path=/20080229/00: list the items which in path 'picture/20080229/00'<br>page=1: list the items in page 1<br>pagesize=20: indicate that there are 20 items in each page<br>And the server would response the fist 20 items which in path 'picture/20080229/00' |
| pagesize   | integer                                  | How many items (file or folder) in a page. See more details in previous parameter.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| total_file | integer                                  | Total number of these items (file or folder)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| total_page | integer                                  | Total page of these items (file or folder)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| num        | integer                                  | Number of items (file or folder) in indicated page                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| items      | string                                   | <b>Attributes of listed items</b><br><name> <type> <recording type> <size or num>:<br><br><b>where:</b><br>name: name of file or folder.<br>type: indicate this 'name' is a file or a folder, f(file) or d(folder);<br>recording type: d (digital input) or m (motion) or n (normal)<br>size or num: file's size(type=f), number of files in the folder(type=d).<br><br><b>Use ' ' (pipe or vertical bar) to separate each of attribute. Use ':' (column) to divide two items.</b><br><br><b>Example:</b><br>items=20090826 d n 100:20090826_000000.avi f n 512:20090825 d n 50:...                                                                                                                                        |

### 3.7.4.download files of SD card

request:

GET /config/sdcard\_download.cgi

parameters:

| Name | Value          | Description                                                                                                                   |
|------|----------------|-------------------------------------------------------------------------------------------------------------------------------|
| type | picture, video | Which type of files would like to see                                                                                         |
| path | string         | Path of file                                                                                                                  |
| file | string         | The file name could be got in command /config/sdcard_list.cgi.<br>The attribute type must be 'f' (only file can be download). |

response (when file is available):

```
HTTP/1.0 200 OK<CRLF>
Content-Type: application/octet-stream<CRLF>
Content-Length: <size of file><CRLF>
<CRLF>
<Binary data of file>
```

response (when file is not available for download):

| Name   | Value   | Description                                                                                                                                   |
|--------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| path   | string  | Path of file                                                                                                                                  |
| file   | string  | The name of file which is wanted to be download.                                                                                              |
| result | integer | The status of download action of indicated file.<br>1: file is not exists<br>2: the 'file' is a folder, it must be a file.(can't be download) |

### 3.7.5.delete files of SD card

request:

GET /config/sdcard\_delete.cgi

parameters:

| Name | Value                    | Description                                                                                                                              |
|------|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| type | picture, video           | Which type of files would like to be deleted                                                                                             |
| path | string                   | Path of file                                                                                                                             |
| name | <file 1>[:<file 2>: ...] | File list which would be deleted. The file name could be get in command /config/sdcard_list.cgi. Use ':'(column) as the split character. |

response:

| Name      | Value                           | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-----------|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| num       | integer                         | Number of items (file or folder) which in 'name' (deleting list).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| path      | string                          | Path of file                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| sd_status | ready,<br>invalid,<br>protected | Status of SD Card. When "invalid" and "protected" is given, which mean the delete is completed failure.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| items     | string                          | <p><b>Attributes of deleted items</b><br/>&lt;name&gt; &lt;type&gt; &lt;status&gt;:</p> <p><b>where:</b><br/>name: name of file or folder.<br/>type: indicate this 'name' is a file or a folder, f(file) or d(folder) or n(unknown);<br/>status: the status of deleting action of indicated item (file or folder), &lt;filename&gt;.<br/>0: item is successfully deleted<br/>1: item does not exist<br/>2: item is not deleted</p> <p><b>Use ' ' (pipe or vertical bar) to separate each of attribute. Use ':' (column) to divide two items.</b></p> <p><b>Example:</b><br/>items=20090826 d 0:20090826_000000.avi n 1: 20090826_101200.avi  f 2:...</p> <p><b>where:</b><br/>20090826: this item is a directory and has been deleted successfully;<br/>20090826_000000.avi: this item is not exists;<br/>20090826_101200.avi: this item is a file and has not been deleted.</p> |



## 4.Streaming

---

### 4.1.Live streaming URL

#### 4.1.1.get a JPEG image

Returns a JPEG image with the default resolution and compression as defined in the configuration.

request:

```
GET /image/jpeg.cgi
```

response:

```
HTTP/1.0 200 OK\r\n
Content-Type: image/jpeg\r\n
Content-Length: <image size>\r\n
\r\n
<JPEG image data>\r\n
```

#### 4.1.2.get motion JPEG video stream

Returns a multipart image stream with the default resolution and compression as defined in the configuration. The content type is "multipart/x-mixed-replace" and each image ends with a boundary string <boundary>.

request:

```
GET /video/mjpg.cgi
```

parameters:

| Name      | Value | Description                                                                                                 |
|-----------|-------|-------------------------------------------------------------------------------------------------------------|
| profileid | #     | optional. If omitted, the url will output one of stream profile that match the assign format (motion JPEG). |

response:

```
HTTP/1.0 200 OK\r\n
Content-Type: multipart/x-mixed-replace;boundary=<boundary>\r\n
\r\n
--<boundary>\r\n
Content-Type: image/jpeg\r\n
Content-Length: <image size>\r\n
\r\n
<JPEG image data>\r\n
--<boundary>\r\n
Content-Type: image/jpeg\r\n
Content-Length: <image size>\r\n
\r\n
<JPEG image data>\r\n
--<boundary>\r\n
```

#### 4.1.3.get MPEG-4 elementary video stream

Returns a MPEG-4 elementary stream with assigned profile id defined in the configuration. The content type is "video/MP4V-ES" please refer to INAN MIME Media Types.

request:

```
GET /video/MP4V-ES.cgi
```

parameters:

| Name      | Value | Description                                                                                                             |
|-----------|-------|-------------------------------------------------------------------------------------------------------------------------|
| profileid | #     | optional. If omitted, the url will output one of stream profile that match the assign format (MPEG4 elementary stream). |

```
response:
HTTP/1.0 200 OK\r\n
Content-Type: video/MP4V-ES\r\n
\r\n
<MPEG-4 ISO/IEC 14496-2 elementary stream>
```

### 4.1.4.get MPEG-4 video stream

Return the MPEG-4 video stream. The video data header please refer to the ACS Stream Header.

request:

```
GET /video/ACVS.cgi
```

parameters:

| Name      | Value | Description                                                                                                       |
|-----------|-------|-------------------------------------------------------------------------------------------------------------------|
| profileid | #     | optional. If omitted, the url will output one of stream profile that match the assign format (ACVS/MPEG4 stream). |

response:

```
HTTP/1.0 200 OK\r\n
Content-type: video/ACVS\r\n
\r\n
<ACAS Video Stream Data>
```

Where <ACAS Video Stream Data> is defined as below:

```
<ACS_VideoHeader>
<MPEG4 Raw Data>
<ACS_AudioHeader>
<MPEG4 Raw Data>
...
```

<ACS\_VideoHeader> is defined in **8.2 Advanced ip-Camera Stream(ACS) Header**.

<MPEG4 Raw Data> is raw data of MPEG4 video stream.

### 4.1.5.get audio stream

The audio data header please refer to the ACS Stream Header.

request:

```
GET /audio/ACAS.cgi
```

parameters:

| Name      | Value | Description                                                                                                     |
|-----------|-------|-----------------------------------------------------------------------------------------------------------------|
| profileid | #     | optional. If omitted, the url will output one of stream profile that match the assign format (ACAS/PCM stream). |

response:

```
HTTP/1.0 200 OK\r\n
Content-type: audio/ACAS\r\n
\r\n
<ACAS Audio Stream Data>
```

Where <ACAS Audio Stream Data> is defined as below:

```
<ACS_AudioHeader>
<Audio Raw Data>
<ACS_AudioHeader>
<Audio Raw Data>
...
```

<ACS\_AudioHeader> is defined in **8.2 Advanced ip-Camera Stream(ACS) Header**.

<Audio Raw Data> is raw data of audio stream. The format of this data depends on <ACS\_AudioHeader>.

### 4.1.6.get profile video stream

Return the video stream associated with a specific profile. The video stream format depends on the compression type of video in

that profile. Please read note below.

request:

GET /video/video.cgi

parameters:

| Name      | Value                               | Description                                                                     |
|-----------|-------------------------------------|---------------------------------------------------------------------------------|
| profileid | # (1 to the count of video profile) | If omitted, the url will output the stream of default profile (profile id = 1). |

response:

```
HTTP/1.0 200 OK\r\n
Content-type: video/ACVS\r\n
\r\n
<video streaming data>
```

note:

<video streaming data>:

If the compression type of the designated profile is motion-JPEG(MJPEG), the stream format is different with as multipart format stream. The format of video stream with profile motion-JPEG is wrapped by ACVS header.

On the other hand, if the compression type is MPEG4 (or H.264 or other advanced compression methods), the output format follows the standard of ACVS format (See Appendix 8.2).

This url is only available while the value of item 'vprofileformat' in 3.3.1 or 3.3.2 is equal to or greater than 1.0

### 4.1.7.put audio upstream (two-way audio talk)

There are two requests to use this service. One of the requests is "verification request", the other is called "uploading request".

While uploading audio data from client to camera server, the client may run into some situations instead of successfully keeping uploading audio data. For example, if another client has been uploading audio data, server will disconnect the connection after client starting uploading audio. On the other hand, if client send command with wrong authentication information, the server will also reject the request from client. So a client should use verification request to test if it has the correct authentication information before uploading audio stream. In other word, if a user has past the verification request but it still got disconnecting after uploading request because of the other connect existing.

#### Verification request:

Request object:

HEAD /dev/speaker.cgi?client=<MAC address of client side>

Request header:

```
Authorization: Basic <base64 encode(username:password)>\r\n
Content-Type: audio/ACAS\r\n
Content-length: 0\r\n
\r\n
```

Response of verification request:

If the authorization is verified, the camera should return 200 OK to indicate client side can keep uploading request:

```
HTTP/1.0 200 OK\r\n
```

If the authorization is failed, the camera would return HTTP error code to indicate client side should stop the uploading request, for example:

```
HTTP/1.0 401 Unauthorized\r\n
```

#### Uploading request:

Request object:

POST /dev/speaker.cgi?client=<MAC address of client side>

Request header:

```
Authorization: Basic <base64 encode(username:password)>\r\n
Content-Type: audio/ACAS\r\n
Content-length: 4\r\n
Connection: Keep-Alive\r\n
\r\n
```

Request body:

<Random 4CC>

(wait for 2 sec.)

```
<AAH>
<1K audio data>
<AAH>
<1K audio data>
<AAH>
<1K audio data>
...
```

**Where:**

<Random 4CC>: 4-byte random char code.

(wait for 2 sec.): After 4-bytes random char code, client should wait for 1 sec before sending more audio data.

<AAH>: the header of AAH defined as follow:

```
typedef struct _ACS_AudioHeader
{
    unsigned long ulHdrID;           //Header ID = 0xF6010000
    unsigned long ulHdrLength;       // sizeof(ACS_AudioHeader)
    unsigned long ulDataLength;      // audio data length
    unsigned long ulSequenceNumber; // sequence number
    unsigned long ulTimeSec;         //sample time stamp
    unsigned long ulTimeUsec;       // sample time stamp
    unsigned long ulDataChecksum;    // not used..
    unsigned short usFormat;         // 0x00000010 S16 LE
    unsigned short usChannels;       // 1 channel
    unsigned short usSampleRate;     // 8000 hz
    unsigned short usSampleBits;     // 16 bits
    unsigned long ulReserved;        //
}ACS_AudioHeader, *PACS_AudioHeader;
```

<1K audio data>: audio data acquired by client side in the format specified by <AAH> header

Response of uploading request:

There are no response for this request.

## 4.1.8.get H.264 video stream

Return the H.264 video stream. The video data header please refer to the ACS Stream Header.

request:

GET /video/ACVS-H264.cgi

parameters:

| Name      | Value | Description                                                                                                       |
|-----------|-------|-------------------------------------------------------------------------------------------------------------------|
| profileid | #     | optional. If omitted, the url will output one of stream profile that match the assign format (ACVS/H.264 stream). |

response:

```
HTTP/1.0 200 OK\r\n
Content-type: video/ACVS\r\n
\r\n
<ACAS Video Stream Data>
```

Where <ACAS Video Stream Data> is defined as below:

```
<ACS_VideoHeader>
<H.264 Raw Data>
<ACS_AudioHeader>
<H.264 Raw Data>
...
```

<ACS\_VideoHeader> is defined in **8.2 Advanced ip-Camera Stream(ACS) Header**.

<MPEG4 Raw Data> is raw data of MPEG4 video stream.

### 4.1.9.get audio WAVE stream

Return the audio stream in WAV format.

request:

GET /audio/x-wav.cgi

parameters:

| Name | Value | Description                                                                                            |
|------|-------|--------------------------------------------------------------------------------------------------------|
| sec  | #     | Duration of audio streaming.<br>0: (default) indicate maximum duration<br>1-120000: duration in second |

response:

```
HTTP/1.0 200 OK\r\n
Content-type: audio/x-wav\r\n
\r\n
<wave format data>
```

Where <wave format data> is a standard Microsoft wave file format. Please refer to MIME: audio/x-wav.

### 4.1.10.get audio MS-ADPCM stream

Return the audio stream in MS-ADPCM format.

request:

GET /audio/ACAS-MSADPCM.cgi

response:

```
HTTP/1.0 200 OK\r\n
Content-type: audio/ACAS\r\n
\r\n
<ACAS Audio Stream Data>
```

Where <ACAS Audio Stream Data> is defined as below:

```
<ACS_AudioHeader>
<Audio Raw Data>
<ACS_AudioHeader>
<Audio Raw Data>
```

...

<ACS\_AudioHeader> is defined in **8.2 Advanced ip-Camera Stream(ACS) Header**.

<Audio Raw Data> is raw data of audio stream. The format of this data depends on <ACS\_AudioHeader>.

### 4.1.11.get audio MU-LAW stream

Return the audio stream in MU-LAW format.

request:

GET /audio/ACAS-ULAW.cgi

response:

```
HTTP/1.0 200 OK\r\n
Content-type: audio/ACAS\r\n
\r\n
<ACAS Audio Stream Data>
```

Where <ACAS Audio Stream Data> is defined as below:

```
<ACS_AudioHeader>
<Audio Raw Data>
<ACS_AudioHeader>
<Audio Raw Data>
```

...

<ACS\_AudioHeader> is defined in **8.2 Advanced ip-Camera Stream(ACS) Header**.

<Audio Raw Data> is raw data of audio stream. The format of this data depends on <ACS\_AudioHeader>.

## 4.1.12.get audio AAC stream

Return the audio stream in AAC format.

request:

```
GET /audio/ACAS- AAC.cgi
```

response:

```
HTTP/1.0 200 OK\r\n
Content-type: audio/ACAS\r\n
\r\n
<ACAS Audio Stream Data>
```

Where <ACAS Audio Stream Data> is defined as below:

<ACS\_AudioHeader>

<Audio Raw Data>

<ACS\_AudioHeader>

<Audio Raw Data>

...

<ACS\_AudioHeader> is defined in **8.2 Advanced ip-Camera Stream(ACS) Header**.

<Audio Raw Data> is raw data of audio stream. The format of this data depends on <ACS\_AudioHeader>.

## 4.1.13.get audio A-LAW stream

Return the audio stream in A-LAW format.

request:

```
GET /audio/ACAS- ALAW.cgi
```

response:

```
HTTP/1.0 200 OK\r\n
Content-type: audio/ACAS\r\n
\r\n
<ACAS Audio Stream Data>
```

Where <ACAS Audio Stream Data> is defined as below:

<ACS\_AudioHeader>

<Audio Raw Data>

<ACS\_AudioHeader>

<Audio Raw Data>

...

<ACS\_AudioHeader> is defined in **8.2 Advanced ip-Camera Stream(ACS) Header**.

<Audio Raw Data> is raw data of audio stream. The format of this data depends on <ACS\_AudioHeader>.

## 4.1.14.get profile audio stream

Return the audio stream associated with a specific profile. The audio stream format depends on the compression type of audio in that profile. Please read note below.

request:

```
GET /audio/audio.cgi
```

parameters:

| Name      | Value                               | Description                                                                     |
|-----------|-------------------------------------|---------------------------------------------------------------------------------|
| profileid | # (1 to the count of audio profile) | If omitted, the url will output the stream of default profile (profile id = 1). |

response:

```
HTTP/1.0 200 OK\r\n
Content-type: audio/ACAS\r\n
\r\n
<audio streaming data>
```

note:

<audio streaming data>:

The audio stream is wrapped by ACAS header.

## 5.Camera Control API

---

### 5.1.Remote control

#### 5.1.1.query PTZ information

request:

GET /config/ptz\_info.cgi

GET /ptz/ptz\_info.cgi (accessible by PTZ privilege group, see 5.1.15)

response: (only supported parameters are displayed.)

| Name           | Value      | Description                                                                                  |
|----------------|------------|----------------------------------------------------------------------------------------------|
| pmax           | An integer | maximum position of pan in degree                                                            |
| pmin           | An integer | minimum position of pan in degree                                                            |
| tmax           | An integer | maximum position of tilt in degree                                                           |
| tmin           | An integer | minimum position of tilt in degree                                                           |
| zmax           | An integer | maximum position of zoom                                                                     |
| zmin           | An integer | minimum position of zoom                                                                     |
| customizedhome | no, yes    | to indicate whether camera can use “customized home” function. Please refer to section 5.1.9 |

#### 5.1.2.get the current PTZ position

request:

GET /config/ptz\_pos.cgi

GET /ptz/ptz\_pos.cgi (accessible by PTZ privilege group, see 5.1.15)

response: (only supported parameters are displayed.)

| Name | Value      | Description       |
|------|------------|-------------------|
| p    | An integer | the pan position  |
| t    | An integer | the tilt position |
| z    | An integer | the zoom position |

#### 5.1.3.get the PTZ movement size in a step

request:

GET /config/ptz\_step.cgi

GET /ptz/ptz\_step.cgi (accessible by PTZ privilege group, see 5.1.15)

response: (only supported parameters are displayed.)

| Name  | Value      | Description                  |
|-------|------------|------------------------------|
| pstep | An integer | pan movement size in a step  |
| tstep | An integer | tilt movement size in a step |
| zstep | An integer | zoom movement size in a step |

#### 5.1.4.set the PTZ movement size in a step

You can specify any of the parameters you want to set.

request:

GET/POST /config/ptz\_step.cgi

parameters:

see the above table.

response:  
see the above table.

### 5.1.5.list all PTZ presets

request:  
GET /config/ptz\_preset\_list.cgi  
GET /ptz/ptz\_preset\_list.cgi (accessible by PTZ privilege group, see 5.1.15)

response:

| Name                  | Value               | Description                                                                                    |
|-----------------------|---------------------|------------------------------------------------------------------------------------------------|
| presets               | <preset name1>, ... | all presets                                                                                    |
| <preset name1><br>... | <p>,<t>,<z><br>...  | the position of the preset name line by line.<br>for example,<br>door1=100,0<br>gate1=-20,-100 |

### 5.1.6.add, delete or goto a PTZ preset

request:  
GET/POST /config/ptz\_preset.cgi  
GET/POST /ptz/ptz\_preset.cgi (accessible by PTZ privilege group, see 5.1.15)

parameters:

| Name | Value            | Description                                                                                                      |
|------|------------------|------------------------------------------------------------------------------------------------------------------|
| name |                  | preset name                                                                                                      |
| act  | add<br>del<br>go | add the current position to the preset<br>delete the preset (only available at /config/ dir)<br>go to the preset |

response:  
see the above table.

### 5.1.7.move PTZ absolutely

request:  
GET/POST /config/ptz\_move.cgi  
GET/POST /ptz/ptz\_move.cgi (accessible by PTZ privilege group, see 5.1.15)

parameters:

| Name | Value      | Description                                        |
|------|------------|----------------------------------------------------|
| p    | An integer | Pans the device relative to the (0,0,0) position.  |
| t    | An integer | Tilts the device relative to the (0,0,0) position. |
| z    | An integer | Zooms the device relative to the (0,0,0) position. |

response:  
see the above table. If the movement is out of boundary, you will get the actual absolute position.

### 5.1.8.move PTZ relatively

request:  
GET/POST /config/ptz\_move\_rel.cgi  
GET/POST /ptz/ptz\_move\_rel.cgi (accessible by PTZ privilege group, see 5.1.15)

parameters:

| Name | Value      | Description                                        |
|------|------------|----------------------------------------------------|
| p    | -32 ... 32 | Pans the device relative to the current position.  |
| t    | -32 ... 32 | Tilts the device relative to the current position. |



|   |            |                                                    |
|---|------------|----------------------------------------------------|
| z | -32 ... 32 | Zooms the device relative to the current position. |
|---|------------|----------------------------------------------------|

response:

see the above table. If the movement is out of boundary, you will get the actual relative p, t, z values it moves.

### 5.1.9.get, set, goto, reset PTZ customized home position

request:

GET/POST /config/ptz\_home.cgi

GET/POST /ptz/ptz\_home.cgi (accessible by PTZ privilege group, see 5.1.15)

parameters:

| Name | Value                     | Description                                                                                                                                                                                                           |
|------|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| act  | get<br>set<br>go<br>reset | get current home position. This is the default value.<br>set current home position (only available at /config/ dir)<br>go to home position<br>reset home position to factory setting (only available at /config/ dir) |
| p    | An integer                | (only for act=set)Pans the device relative to the default (0,0,0) position.                                                                                                                                           |
| t    | An integer                | (only for act=set)Tilts the device relative to the default (0,0,0) position.                                                                                                                                          |
| z    | An integer                | (only for act=set)Zooms the device relative to the default (0,0,0) position.                                                                                                                                          |

response:

Return current home position.

| Name | Value      | Description       |
|------|------------|-------------------|
| p    | An integer | the pan position  |
| t    | An integer | the tilt position |
| z    | An integer | the zoom position |

note:

If no any parameter is given, the effect will equivalent giving 'act=get'. If any of parameters p,t,z is given and the value of parameter 'act' is not 'set', it will be ignored by camera.

### 5.1.10.Auto Patrol

request:

GET/POST /config/auto\_patrol.cgi

GET/POST /ptz/auto\_patrol.cgi (accessible by PTZ privilege group, see 5.1.15)

parameters:

| Name | Value                      | Description                                                                                                                                             |
|------|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| act  | go<br>[continue]<br>[stop] | Run PTZ's auto patrol.<br>act=go means run auto patrol function once<br>act=[continue] or act=[stop] means begin continuous patrol mode and stop patrol |

response:

see the above table.

note:

The item enclosed by [ ] means optional value. That optional value can be used only in some special models.

### 5.1.11.Auto Pan

request:

GET/POST /config/auto\_pan.cgi

GET/POST /ptz/auto\_pan.cgi (accessible by PTZ privilege group, see 5.1.15)

parameters:

| Name | Value | Description |
|------|-------|-------------|
|------|-------|-------------|

|     |                            |                                                                                                                                             |
|-----|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| act | go<br>[continue]<br>[stop] | Run PTZ's auto pan.<br>act=go means run auto pan function once<br>act=[continue] or act=[stop] means begin continuous pan mode and stop pan |
|-----|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|

response:  
see the above table.

note:  
The item enclosed by [ ] means optional value. That optional value can be used only in some special models.

### 5.1.12.Configure sequence order of presets for Auto Patrol

request:  
GET /config/config\_auto\_patrol.cgi

parameters:

| Name    | Value                              | Description                                                                                                                                                                                                               |
|---------|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| presets | <preset name1>, <preset name2>,... | A sequence of presets. The CGI auto_patrol function move camera PTZ preset by this sequence. Maximum count of preset in this sequence is 20. Note: if this parameter is not given, the camera will list current sequence. |

response:  
see the above table. If the count of given preset point is greater than 20, only first 20 preset in the sequence will kept by camera.

### 5.1.13.get, set the type of focus function: auto focus or manual focus

request:  
GET/POST /config/focus\_type.cgi  
GET/POST /ptz/focus\_type.cgi

parameters:

| Name | Value      | Description                                                                                 |
|------|------------|---------------------------------------------------------------------------------------------|
| act  | get<br>set | get current focus type.<br>set the focus type                                               |
| type | An integer | (only for act=set) the type of focus function<br>type=1: auto focus<br>type=0: manual focus |

response:

| Name | Value      | Description                                                              |
|------|------------|--------------------------------------------------------------------------|
| type | An integer | the type of focus function<br>type=1: auto focus<br>type=0: manual focus |

### 5.1.14.adjust the focus manually, focus near or focus far from current position

request:  
GET/POST /config/manual\_focus.cgi  
GET/POST /ptz/manual\_focus.cgi

parameters:

| Name | Value                 | Description                                                                                                                                                     |
|------|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| step | An integer (-32...32) | The step for focus near or focus far. The positive value means focus far (focus on distant object), the negative value means focus near (focus on near object). |

response:  
See the above table. If the movement is out of boundary, you will get the actual step performed. If current focus type is not manual focus, this CGI does nothing and returns "step=0"

### 5.1.15.get PTZ control privilege groups

This command allows you to query the list of privilege groups which could access /ptz/ directory to control PTZ and focus of camera.

request:

GET /config/ptz\_privilege.cgi

response:

| Name   | Value    | Description                                                                                                                                                                                                                                |
|--------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| groups | A string | List of groups separated by comma which are given the privilege to access /ptz/ directory to control PTZ and focus of camera. Administrator group are always accessible to /ptz/ directory, not matter is listed in this parameter or not. |

### 5.1.16.set PTZ control privilege groups

This command allows you to configure the list of privilege groups which could access to /ptz/ directory to control PTZ and focus of camera.

request:

GET/POST /config/ptz\_privilege.cgi

parameters:

See the above table.

response:

See the above table.

### 5.1.17.query focus information

request:

GET /config/focus\_info.cgi

response:

| Name | Value      | Description                |
|------|------------|----------------------------|
| max  | An integer | maximum position of focus. |
| min  | An integer | minimum position of focus. |

### 5.1.18.get the current position of focus

request:

GET /config/focus\_pos.cgi

response:

| Name  | Value      | Description         |
|-------|------------|---------------------|
| focus | An integer | The focus position. |

### 5.1.19.set absolutely position of focus

request:

GET/POST /config/focus\_pos.cgi

parameters:

see the above table.

response:

see the above table.

### 5.1.20.fine-tune focus automatically

request:

GET/POST /config/focus\_act.cgi

parameters:

| Name      | Value | Description                    |
|-----------|-------|--------------------------------|
| autofocus | yes   | Fine-tune focus automatically. |

response:

| Name      | Value     | Description                         |
|-----------|-----------|-------------------------------------|
| autofocus | yes, fail | The action was successful or fails. |

## 5.1.21.PTZ direction of movement

request:

GET/POST /config/ptz\_direction.cgi

parameters:

| Name      | Value      | Description                                                                                                          |
|-----------|------------|----------------------------------------------------------------------------------------------------------------------|
| direction | String     | Use<br>“up”、“down”、“left”、“right”、“upleft”、“upright”、“downleft”、“downright”、“stop”、“zoomwide”、“zoomtele”、“zoomstop”. |
| speed     | An integer | speed control for PTZ (1 – 10)                                                                                       |

## 5.2.via RS-485

### 5.2.1.do RS-485 commands

request:

GET/POST /config/rs485\_do.cgi

parameters:

| Name      | Value      | Description                                                                                                                                                                                                                                                                                                                                     |
|-----------|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| direction | 0-13       | 0: wide (zoom out) with stop.<br>1: up with stop<br>2: tele (zoom in) with stop<br>3: left with stop<br>4: home (only for custom command)<br>5: right with stop<br>6: focus far with stop<br>7: down with stop<br>8: focus near with stop<br>9:<br>10: custom command 1<br>11: custom command 2<br>12: custom command 3<br>13: custom command 4 |
| speed     | An integer | speed control for up, down, left, right. (1 – 10)<br>(includes: Dyna, Lilin, Lilin2, PelcoD, PelcoP)                                                                                                                                                                                                                                            |

## 5.3.PTDC Pan/Tilt get information

### 5.3.1.get Pan/Tilt Position

Get the current Pan/Tilt position degree

request:

GET /cgi/ptdc.cgi?command=get\_pos

|                                 |                                                                                                     |
|---------------------------------|-----------------------------------------------------------------------------------------------------|
| Response:<br>Represented by XML | <pre>&lt;config&gt;   &lt;posX&gt;0&lt;/posX&gt;   &lt;posY&gt;0&lt;/posY&gt; &lt;/config&gt;</pre> |
|---------------------------------|-----------------------------------------------------------------------------------------------------|

### 5.3.2.get Pan/Tilt Position by Step

Get the current Pan/Tilt position step

request:

GET /cgi/ptdc.cgi?command=get\_pos\_step

|                                 |                                                                                                                     |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------|
| Response:<br>Represented by XML | <pre>&lt;config&gt;   &lt;posXStep&gt;0&lt;/posXStep&gt;   &lt;posYStep&gt;0&lt;/posYStep&gt; &lt;/config&gt;</pre> |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------|

### 5.3.3.get Pan/Tilt Boundary

Get Pan/Tilt position degree boundary

request:

GET /cgi/ptdc.cgi?command=get\_boundary

|                                 |                                                                                                                                                                                                                                       |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Response:<br>Represented by XML | <pre>&lt;config&gt;   &lt;boundaryMaxX&gt;164&lt;/boundaryMaxX&gt;   &lt;boundaryMinX&gt;-164&lt;/boundaryMinX&gt;   &lt;boundaryMaxY&gt;83&lt;/boundaryMaxY&gt;   &lt;boundaryMinY&gt;-30&lt;/boundaryMinY&gt; &lt;/config&gt;</pre> |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### 5.3.4.get Pan/Tilt Boundary by Step

Get Pan/Tilt position step boundary

request:

GET /cgi/ptdc.cgi?command=get\_step\_boundary

|                                 |                                                                                                                                                                                                                                                                                |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Response:<br>Represented by XML | <pre>&lt;config&gt;   &lt;boundaryMaxStepX&gt;18944&lt;/boundaryMaxStepX&gt;   &lt;boundaryMinStepX&gt;-18944&lt;/boundaryMinStepX&gt;   &lt;boundaryMaxStepY&gt;14528&lt;/boundaryMaxStepY&gt;   &lt;boundaryMinStepY&gt;-5360&lt;/boundaryMinStepY&gt; &lt;/config&gt;</pre> |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### 5.3.5.get Pan/Tilt Accuracy

Get the current Pan/Tilt minimum valid movement and precision degree.

request:

GET /cgi/ptdc.cgi?command=get\_pt\_accuracy

|                                 |                                                                                                                                                                                                                                                       |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Response:<br>Represented by XML | <pre>&lt;config&gt;   &lt;panMinMovement&gt;0.14&lt;/panMinMovement&gt;   &lt;panPrecision&gt;0.14&lt;/panPrecision&gt;   &lt;tiltMinMovement&gt;0.18&lt;/tiltMinMovement&gt;   &lt;tiltPrecision&gt;0.09&lt;/tiltPrecision&gt; &lt;/config&gt;</pre> |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

note:

**xxxMinMovement:** user has to set the minimum degree in every movement via Pan and/or Tilt to make sure the physical PT's movement visible.

**xxxPrecision:** the movement unit, user has to set the degree as multiple times of xxxPrecision for PT hardware precision consideration.

### 5.3.6.get Pan/Tilt Accuracy by Step

Get the current Pan/Tilt minimum valid movement and precision steps.

request:

GET /cgi/ptdc.cgi?command=get\_pt\_step\_accuracy

|                                 |                                                                                                                                                                                                                                                                               |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Response:<br>Represented by XML | <pre>&lt;config&gt;   &lt;panMinMovementStep&gt;16&lt;/panMinMovementStep&gt;   &lt;panPrecisionStep&gt;16&lt;/panPrecisionStep&gt;   &lt;tiltMinMovementStep&gt;32&lt;/tiltMinMovementStep&gt;   &lt;tiltPrecisionStep&gt;16&lt;/tiltPrecisionStep&gt; &lt;/config&gt;</pre> |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

note:

**xxxMinMovementStep:** user has to set the minimum steps in every movement via Pan and/or Tilt to make sure the physical PT's movement visible.

**xxxPrecisionStep:** the movement unit, user has to set the steps as multiple times of PrecisionStep for PT hardware precision consideration.

### 5.3.7.get Pan/Tilt View Angle

Get Pan/Tilt view angle degree in live view.request:

GET /cgi/ptdc.cgi?command=get\_view\_angle

|                                 |                                                                                                                                                               |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Response:<br>Represented by XML | <pre>&lt;config&gt;   &lt;viewAngleHorizontal&gt;51&lt;/viewAngleHorizontal&gt;   &lt;viewAngleVertical&gt;39&lt;/viewAngleVertical&gt; &lt;/config&gt;</pre> |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|

### 5.3.8.get Pan/Tilt View Step

Get Pan/Tilt view angle degree in live view.request:

GET /cgi/ptdc.cgi?command=get\_view\_step

|                                 |                                                                                                                                                               |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Response:<br>Represented by XML | <pre>&lt;config&gt;   &lt;viewStepHorizontal&gt;5865&lt;/viewStepHorizontal&gt;   &lt;viewStepVertical&gt;6747&lt;/viewStepVertical&gt; &lt;/config&gt;</pre> |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|

### 5.3.9.get Pan/Tilt Preset Positions

Get the current Pan/tilt Preset Positions' information:

GET /cgi/ptdc.cgi?command=get\_preset\_positions

|                                 |                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Response:<br>Represented by XML | <pre>&lt;config&gt;   &lt;size&gt;2&lt;/size&gt;   &lt;presetName0&gt;1&lt;/presetName0&gt;   &lt;presetX0&gt;0&lt;/presetX0&gt;   &lt;presetY0&gt;30&lt;/presetY0&gt;   &lt;presetZ0&gt;1&lt;/presetZ0&gt;   &lt;presetF0&gt;0&lt;/presetF0&gt;   &lt;presetFType0&gt;1&lt;/presetFType0&gt;   &lt;presetName1&gt;2&lt;/presetName1&gt;   &lt;presetX1&gt;-60&lt;/presetX1&gt;</pre> |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|  |                                                                                                                                                                                   |
|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <pre> &lt;presetY1&gt;30&lt;/presetY1&gt; &lt;presetZ1&gt;1&lt;/presetZ1&gt; &lt;presetF1&gt;0&lt;/presetF1&gt; &lt;presetFType1&gt;1&lt;/presetFType1&gt; &lt;/config&gt; </pre> |
|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

note:

X,Y Unit: degree (pan/tilt)

Z Unit:optical mag (zoom)

F Unit:Step (focus)

FT 0: manual 1: auto (focus type)

### 5.3.10.get Pan/Tilt/Zoom Hardware information

Get the current Pan/Tilt/Zoom/Focus Hardware information:

GET /cgi/ptdc.cgi?command=get\_ptz\_hw\_info

|                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Response:<br/>Represented by XML</p> | <pre> &lt;config&gt;   &lt;pan&gt;     &lt;inAngle&gt;       &lt;viewAngleHorizontal&gt;45&lt;/viewAngleHorizontal&gt;       &lt;boundaryMaxX&gt;164&lt;/boundaryMaxX&gt;       &lt;boundaryMinX&gt;-164&lt;/boundaryMinX&gt;       &lt;panMinMovement&gt;0.14&lt;/panMinMovement&gt;       &lt;panPrecision&gt;0.14&lt;/panPrecision&gt;     &lt;/inAngle&gt;     &lt;inStep&gt;       &lt;viewStepHorizontal&gt;5175&lt;/viewStepHorizontal&gt;       &lt;boundaryMaxStepX&gt;18944&lt;/boundaryMaxStepX&gt;       &lt;boundaryMinStepX&gt;-18944&lt;/boundaryMinStepX&gt;       &lt;panMinMovementStep&gt;16&lt;/panMinMovementStep&gt;       &lt;panPrecisionStep&gt;16&lt;/panPrecisionStep&gt;     &lt;/inStep&gt;   &lt;/pan&gt;   &lt;tilt&gt;     &lt;inAngle&gt;       &lt;viewAngleVertical&gt;38&lt;/viewAngleVertical&gt;       &lt;boundaryMaxY&gt;83&lt;/boundaryMaxY&gt;       &lt;boundaryMinY&gt;-30&lt;/boundaryMinY&gt;       &lt;tiltMinMovement&gt;0.18&lt;/tiltMinMovement&gt;       &lt;tiltPrecision&gt;0.09&lt;/tiltPrecision&gt;     &lt;/inAngle&gt;     &lt;inStep&gt;       &lt;viewStepVertical&gt;6574&lt;/viewStepVertical&gt;       &lt;boundaryMaxStepY&gt;14528&lt;/boundaryMaxStepY&gt;       &lt;boundaryMinStepY&gt;-5360&lt;/boundaryMinStepY&gt;       &lt;tiltMinMovementStep&gt;32&lt;/tiltMinMovementStep&gt;       &lt;tiltPrecisionStep&gt;16&lt;/tiltPrecisionStep&gt;     &lt;/inStep&gt;   &lt;/tilt&gt;   &lt;zoom&gt;     &lt;inMag&gt;       &lt;boundaryMaxZoom&gt;10&lt;/boundaryMaxZoom&gt;       &lt;boundaryMinZoom&gt;1&lt;/boundaryMinZoom&gt;       &lt;zoomMinMovement&gt;0.09&lt;/ zoomMinMovement&gt;       &lt;zoomPrecision&gt;0.09&lt;/ zoomPrecision&gt;     &lt;/inMag&gt;     &lt;inStep&gt;       &lt;zoomMaxStep&gt;1174&lt;/ zoomMaxStep&gt;       &lt;zoomMinStep&gt;0&lt;/zoomMinStep&gt;       &lt;zoomMinMovementStep&gt;1&lt;/zoomMinMovementStep&gt;       &lt;zoomPrecisionStep&gt;1&lt;/zoomPrecisionStep&gt;     &lt;/inStep&gt;   &lt;/zoom&gt;   &lt;focus&gt;     &lt;boundaryMaxFocus&gt;1038&lt;/boundaryMaxFocus&gt; </pre> |
|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|  |                                                                                                                                                                                                                    |
|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <pre> &lt;boundaryMinFocus&gt;194&lt;/boundaryMinFocus&gt; &lt;focusMinMovementStep&gt;1&lt;/focusMinMovementStep&gt; &lt;focusPrecisionStep&gt;1&lt;/focusPrecisionStep&gt; &lt;/focus&gt; &lt;/config&gt; </pre> |
|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

note:

Pan/Tilt Unit: degree

Zoom :

inMag Unit: optical mag

Zoom :

inStep Unit: step

Focus Unit: step

### 5.3.11.get Pan/Tilt Home Position

Get the current home Pan/Tilt position:

GET /cgi/ptdc.cgi?command=get\_home\_pos

|                                 |                                                                                                                       |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| Response:<br>Represented by XML | <pre> &lt;config&gt;   &lt;homePosX&gt;0&lt;/homePosX&gt;   &lt;homePosY&gt;0&lt;/homePosY&gt; &lt;/config&gt; </pre> |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------|

### 5.3.12.get Pan/Tilt Patrol Speed

Get the patrol speed in patrol action:

GET /cgi/ptdc.cgi?command=get\_patrol\_speed

|                                 |                                                                            |
|---------------------------------|----------------------------------------------------------------------------|
| Response:<br>Represented by XML | <pre> &lt;config&gt;   &lt;speed&gt;5&lt;/speed&gt; &lt;/config&gt; </pre> |
|---------------------------------|----------------------------------------------------------------------------|

### 5.3.13.get Pan/Tilt Wait Time

Get the wait\_time in patrol action:

GET /cgi/ptdc.cgi?command=get\_wait\_time

|                                 |                                                                                  |
|---------------------------------|----------------------------------------------------------------------------------|
| Response:<br>Represented by XML | <pre> &lt;config&gt;   &lt;waitTime&gt;5&lt;/waitTime&gt; &lt;/config&gt; </pre> |
|---------------------------------|----------------------------------------------------------------------------------|

note:

Unit: second

when patrol to a turning point, it will stop and wait for “wait\_time” seconds

## 5.4.PTDC Pan/Tilt set information

### 5.4.1.set Pan/Tilt Position

Set the Pan/Tilt absolute position in degree unit:



request:

GET/POST /cgi/ptdc.cgi?command=set\_pos&posX=100&posY=30

parameters:

| Name | Value | Description                                 |
|------|-------|---------------------------------------------|
| posX | float | The movement degree is based on Zoom mag 1x |
| posY | float | The movement degree is based on Zoom mag 1x |

response:

A result tag will also be presented:

|                               |                                          |                                    |
|-------------------------------|------------------------------------------|------------------------------------|
| Result:<br>Represented by XML | <result><br><code>ok</code><br></result> |                                    |
| Result code                   | 1. ok                                    | Success                            |
|                               | 2. invalidParameter                      | Need correct posX, posY parameters |
|                               | 3. failure                               | Error command type                 |
|                               | 4. oob                                   | Out Of Boundary                    |

### 5.4.2.set Pan/Tilt Relative Position

Set the Pan/Tilt relative position in degree unit:

request:

GET/POST /cgi/ptdc.cgi?command=set\_relative\_pos&posX=100&posY=30

parameters:

| Name | Value | Description                                 |
|------|-------|---------------------------------------------|
| posX | float | The movement degree is based on Zoom mag 1x |
| posY | float | The movement degree is based on Zoom mag 1x |

response:

A result tag will also be presented:

|                               |                                          |                                    |
|-------------------------------|------------------------------------------|------------------------------------|
| Result:<br>Represented by XML | <result><br><code>ok</code><br></result> |                                    |
| Result code                   | 1. ok                                    | Success                            |
|                               | 2. invalidParameter                      | Need correct posX, posY parameters |
|                               | 3. failure                               | Error command type                 |
|                               | 4. oob                                   | Out Of Boundary                    |

### 5.4.3.set Pan/Tilt Position by Step

Set the Pan/Tilt absolute position in step unit:

request:

GET/POST /cgi/ptdc.cgi?command=set\_pos\_step&posXStep=100&posYStep=30

parameters:

| Name     | Value | Description                                          |
|----------|-------|------------------------------------------------------|
| posXStep | #     | The movement step is not need to base on Zoom mag 1x |
| posYStep | #     | The movement step is not need to base on Zoom mag 1x |

response:

A result tag will also be presented:

|                               |                                          |                                            |
|-------------------------------|------------------------------------------|--------------------------------------------|
| Result:<br>Represented by XML | <result><br><code>ok</code><br></result> |                                            |
| Result code                   | 1. ok                                    | Success                                    |
|                               | 2. invalidParameter                      | Need correct posXStep, posYStep parameters |
|                               | 3. failure                               | Error command type                         |
|                               | 4. oob                                   | Out Of Boundary                            |

### 5.4.4.set Pan/Tilt Relative Position by Step

Set the Pan/Tilt relative position in step unit:

request:

GET/POST /cgi/ptdc.cgi?command=set\_relative\_pos\_step&posXStep=100&posYStep=30

parameters:

| Name     | Value | Description                                          |
|----------|-------|------------------------------------------------------|
| posXStep | #     | The movement step is not need to base on Zoom mag 1x |
| posYStep | #     | The movement step is not need to base on Zoom mag 1x |

response:

A result tag will also be presented:

|                               |                                          |                                            |
|-------------------------------|------------------------------------------|--------------------------------------------|
| Result:<br>Represented by XML | <result><br><code>ok</code><br></result> |                                            |
| Result code                   | 1. ok                                    | Success                                    |
|                               | 2. invalidParameter                      | Need correct posXStep, posYStep parameters |
|                               | 3. failure                               | Error command type                         |
|                               | 4. oob                                   | Out Of Boundary                            |

### 5.4.5.set Home

Set the current Pan/Tilt/Zoom/Focus/FocusType as home position:

request:

GET/POST /cgi/ptdc.cgi?command=set\_home

response:

A result tag will also be presented:

|                               |                                          |         |
|-------------------------------|------------------------------------------|---------|
| Result:<br>Represented by XML | <result><br><code>ok</code><br></result> |         |
| Result code                   | 1. ok                                    | Success |

### 5.4.6.Restore Default Home

Restore to the factory default home position:

request:

GET/POST /cgi/ptdc.cgi?command=restore\_home

response:

A result tag will also be presented:

|                               |                                          |         |
|-------------------------------|------------------------------------------|---------|
| Result:<br>Represented by XML | <result><br><code>ok</code><br></result> |         |
| Result code                   | 1. ok                                    | Success |

### 5.4.7.set Patrol Speed

Set the patrol speed(single\_pan/pan\_patrol/single\_patrol/user\_patrol):

request:

GET/POST /cgi/ptdc.cgi?command=set\_patrol\_speed&speed=3

parameters:

| Name  | Value | Description            |
|-------|-------|------------------------|
| speed | #     | Patrol Speed(1,2,3...) |

response:

A result tag will also be presented:

|                               |                                          |                              |
|-------------------------------|------------------------------------------|------------------------------|
| Result:<br>Represented by XML | <result><br><code>ok</code><br></result> |                              |
| Result code                   | 1. ok                                    | Success                      |
|                               | 2. invalidParameter                      | Need correct speed parameter |
|                               | 3. failure                               | Error command type           |
|                               | 4. oob                                   | Out Of Boundary              |

### 5.4.8.set Patrol Wait Time

Set the wait time for patrol action.

request:

GET/POST /cgi/ptdc.cgi?command=set\_wait\_time&waitTime=3

parameters:

| Name     | Value | Description          |
|----------|-------|----------------------|
| waitTime | #     | between 0 ~ 3600 sec |

response:

A result tag will also be presented:

|                               |                                          |                                 |
|-------------------------------|------------------------------------------|---------------------------------|
| Result:<br>Represented by XML | <result><br><code>ok</code><br></result> |                                 |
| Result code                   | 1. ok                                    | Success                         |
|                               | 2. invalidParameter                      | Need correct waitTime parameter |
|                               | 3. failure                               | Error command type              |
|                               | 4. oob                                   | Out Of Boundary                 |

## 5.5.PTDC Pan/Tilt other Parts

### 5.5.1.Calibration

Pan/Tilt calibration, and then move to current home position.

request:

GET/POST /cgi/ptdc.cgi?command=calibration

response:

A result tag will also be presented:

|                               |                                          |         |
|-------------------------------|------------------------------------------|---------|
| Result:<br>Represented by XML | <result><br><code>ok</code><br></result> |         |
| Result code                   | 1. ok                                    | Success |

### 5.5.2.Single Pan

Pan/Tilt patrol along pan direction once.

request:

GET/POST /cgi/ptdc.cgi?command=single\_pan

response:

A result tag will also be presented:

|                               |                                          |         |
|-------------------------------|------------------------------------------|---------|
| Result:<br>Represented by XML | <result><br><code>ok</code><br></result> |         |
| Result code                   | 1. ok                                    | Success |

### 5.5.3.Pan Patrol

Pan/Tilt patrol along pan direction for user define times.

request:

GET/POST /cgi/ptdc.cgi?command=pan\_patrol

response:

A result tag will also be presented:

|                               |                                          |         |
|-------------------------------|------------------------------------------|---------|
| Result:<br>Represented by XML | <result><br><code>ok</code><br></result> |         |
| Result code                   | 1. ok                                    | Success |

### 5.5.4.Single Patrol

Pan/Tilt patrol depend on user define route once.

request:

GET/POST /cgi/ptdc.cgi?command=single\_patrol

response:

A result tag will also be presented:

|                               |                                          |         |
|-------------------------------|------------------------------------------|---------|
| Result:<br>Represented by XML | <result><br><code>ok</code><br></result> |         |
| Result code                   | 1. ok                                    | Success |

### 5.5.5.User Patrol

Pan/Tilt patrol depend on user define route and times.

request:

GET/POST /cgi/ptdc.cgi?command=user\_patrol

response:

A result tag will also be presented:

|                               |                                          |         |
|-------------------------------|------------------------------------------|---------|
| Result:<br>Represented by XML | <result><br><code>ok</code><br></result> |         |
| Result code                   | 1. ok                                    | Success |

### 5.5.6.Stop Patrol

soft stop(it will not stop immediately until reach a turning point)

request:

GET/POST /cgi/ptdc.cgi?command=stop\_patrol

response:

A result tag will also be presented:

|                               |                                          |         |
|-------------------------------|------------------------------------------|---------|
| Result:<br>Represented by XML | <result><br><code>ok</code><br></result> |         |
| Result code                   | 1. ok                                    | Success |

### 5.5.7.Stop

Hard stop (Pan/Tilt will stop immediately)

request:

GET/POST /cgi/ptdc.cgi?command=stop

response:

A result tag will also be presented:

|                               |                                          |         |
|-------------------------------|------------------------------------------|---------|
| Result:<br>Represented by XML | <result><br><code>ok</code><br></result> |         |
| Result code                   | 1. ok                                    | Success |

### 5.5.8.Go Home

Move the Pan/Tilt to home position:

request:

GET/POST /cgi/ptdc.cgi?command=go\_home

response:

A result tag will also be presented:

|                               |                                          |  |
|-------------------------------|------------------------------------------|--|
| Result:<br>Represented by XML | <result><br><code>ok</code><br></result> |  |
|-------------------------------|------------------------------------------|--|

|             |       |         |
|-------------|-------|---------|
| Result code | 1. ok | Success |
|-------------|-------|---------|

### 5.5.9.Goto Preset Position

Go to preset point as user define (index 0 is the first point as user define instead of index1) (zero base)

request:

GET/POST /cgi/ptdc.cgi?command=goto\_preset\_position&index=0

or

GET/POST /cgi/ptdc.cgi?command=goto\_preset\_position&presetName=aaa

response:

A result tag will also be presented:

|                               |                                          |                                            |
|-------------------------------|------------------------------------------|--------------------------------------------|
| Result:<br>Represented by XML | <result><br><code>ok</code><br></result> |                                            |
| Result code                   | 1. ok                                    | Success                                    |
|                               | 2. invalidParameter                      | Need correct index or presetName parameter |
|                               | 3. failure                               | Error command type                         |

## 5.6.PTDC Zoom/Focus/Focus Type get information

### 5.6.1.get Zoom Boundary

Get Zoom mag boundary, 1 represent as 1x mag, 10 represent as 10x mag.

request:

GET /cgi/ptdc.cgi?command=get\_zoom\_boundary

|                                 |                                                                                                        |
|---------------------------------|--------------------------------------------------------------------------------------------------------|
| Response:<br>Represented by XML | <config><br><boundaryMaxZoom>10</boundaryMaxZoom><br><boundaryMinZoom>1</boundaryMinZoom><br></config> |
|---------------------------------|--------------------------------------------------------------------------------------------------------|

note:

Unit: mag

### 5.6.2.get Zoom Mag

Get the current Zoom mag.

request:

GET /cgi/ptdc.cgi?command=get\_zoom\_mag

|                                 |                                                 |
|---------------------------------|-------------------------------------------------|
| Response:<br>Represented by XML | <config><br><zoomMag>5.5</zoomMag><br></config> |
|---------------------------------|-------------------------------------------------|

note:

The upper return value represent the current optical mag is 5.5x

### 5.6.3.get Zoom Boundary by Step

Get Zoom step boundary.

request:

GET /cgi/ptdc.cgi?command=get\_zoom\_step\_boundary

|                                 |                                                                                                                                 |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| Response:<br>Represented by XML | <pre>&lt;config&gt;   &lt;zoomMaxStep&gt;1174&lt;/zoomMaxStep&gt;   &lt;zoomMinStep 0&lt;/zoomMinStep&gt; &lt;/config&gt;</pre> |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------|

#### 5.6.4.get Zoom Step

Get the current Zoom step.

request:

GET /cgi/ptdc.cgi?command=get\_zoom\_step

|                                 |                                                                                   |
|---------------------------------|-----------------------------------------------------------------------------------|
| Response:<br>Represented by XML | <pre>&lt;config&gt;   &lt;zoomStep&gt;1174&lt;/zoomStep&gt; &lt;/config&gt;</pre> |
|---------------------------------|-----------------------------------------------------------------------------------|

note:

Unit: step

The upper return value represent the current optical step is 1174

#### 5.6.5.get Focus Boundary

Get Focus boundary.

request:

GET /cgi/ptdc.cgi?command=get\_focus\_boundary

|                                 |                                                                                                                                                          |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Response:<br>Represented by XML | <pre>&lt;config&gt;   &lt;boundaryMaxFocus&gt;1024&lt;/boundaryMaxFocus&gt;   &lt;boundaryMinFocus&gt;194&lt;/boundaryMinFocus&gt; &lt;/config&gt;</pre> |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|

note:

Unit: step

The upper return value represent the focus step range can be set.

#### 5.6.6.get Focus Step

Get the current Focus step.

request:

GET /cgi/ptdc.cgi?command=get\_focus\_step

|                                 |                                                                                     |
|---------------------------------|-------------------------------------------------------------------------------------|
| Response:<br>Represented by XML | <pre>&lt;config&gt;   &lt;focusStep&gt;1000&lt;/focusStep&gt; &lt;/config&gt;</pre> |
|---------------------------------|-------------------------------------------------------------------------------------|

#### 5.6.7.get Focus Type

Get the current Focus Type.

request:

GET /cgi/ptdc.cgi?command=get\_zoom\_boundary

|                                 |                                                                                  |
|---------------------------------|----------------------------------------------------------------------------------|
| Response:<br>Represented by XML | <pre>&lt;config&gt;   &lt;focusType&gt;1&lt;/focusType&gt; &lt;/config&gt;</pre> |
|---------------------------------|----------------------------------------------------------------------------------|

note:

focusType 0:manual focus 1:auto focus

### 5.6.8.get Zoom Accuracy

Get the current Zoom minimum valid movement and precision mag.

request:

GET /cgi/ptdc.cgi?command=get\_zoom\_accuracy

|                                 |                                                                                                         |
|---------------------------------|---------------------------------------------------------------------------------------------------------|
| Response:<br>Represented by XML | <config><br><zoomMinMovement>0.09</zoomMinMovement><br><zoomPrecision>0.09</zoomPrecision><br></config> |
|---------------------------------|---------------------------------------------------------------------------------------------------------|

note:

Unit: mag

**xxxMinMovement**: user has to set the minimum mag in every movement via Zoom to make sure the physical Zoom's movement visible.

**xxxPrecision**: the movement unit, user has to set the mag as multiple times of xxxPrecision for Zoom hardware precision consideration.

### 5.6.9.get Focus Type

Get the current Zoom minimum valid movement and precision steps.

request:

GET /cgi/ptdc.cgi?command=get\_zoom\_step\_accuracy

|                                 |                                                                                                                   |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------|
| Response:<br>Represented by XML | <config><br><zoomMinMovementStep>1</zoomMinMovementStep><br><zoomPrecisionStep>1</zoomPrecisionStep><br></config> |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------|

note:

Unit: step

**xxxMinMovementStep**: user has to set the minimum steps in every movement via Zoom to make sure the physical Zoom's movement visible.

**xxxPrecisionStep**: the movement unit, user has to set the steps as multiple times of PrecisionStep for Zoom hardware precision consideration.

### 5.6.10.get Focus Accuracy

Get the current focus minimum valid movement and precision steps.

request:

GET /cgi/ptdc.cgi?command=get\_focus\_accuracy

|                                 |                                                                                                                       |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| Response:<br>Represented by XML | <config><br><focusMinMovementStep>1</focusMinMovementStep><br><focusPrecisionStep>1</focusPrecisionStep><br></config> |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------|

note:

Unit: step

**xxxMinMovementStep**: user has to set the minimum steps in every movement via Focus to make sure the physical Focus's movement visible.

**xxxPrecisionStep**: the movement unit, user has to set the steps as multiple times of PrecisionStep for Focus hardware precision consideration.

### 5.6.11.get Home Zoom Mag

Get the current home Zoom mag.



request:

GET /cgi/ptdc.cgi?command=get\_home\_zoom\_mag

|                                 |                                                         |
|---------------------------------|---------------------------------------------------------|
| Response:<br>Represented by XML | <config><br><homeZoomMag>5.5</homeZoomMag><br></config> |
|---------------------------------|---------------------------------------------------------|

note:

Unit: mag

The upper return value represent the home optical mag is 5.5x

## 5.6.12.get Home Focus Step

Get the current home Zoom step.

request:

GET /cgi/ptdc.cgi?command=get\_home\_zoom\_step

|                                 |                                                            |
|---------------------------------|------------------------------------------------------------|
| Response:<br>Represented by XML | <config><br><homeZoomStep>1174</homeZoomStep><br></config> |
|---------------------------------|------------------------------------------------------------|

note:

Unit: step

The following return value represent the home optical step is 1174

## 5.6.13.get Home Focus Step

Get the current home focus step.

request:

GET /cgi/ptdc.cgi?command=get\_home\_focus\_step

|                                 |                                                              |
|---------------------------------|--------------------------------------------------------------|
| Response:<br>Represented by XML | <config><br><homeFocusStep>1000</homeFocusStep><br></config> |
|---------------------------------|--------------------------------------------------------------|

note:

Unit: step

The following return value represent the home focus step is 1000

## 5.6.14.get Home Focus Type

Get the current home focus type

request:

GET /cgi/ptdc.cgi?command=get\_home\_focus\_type

|                                 |                                                           |
|---------------------------------|-----------------------------------------------------------|
| Response:<br>Represented by XML | <config><br><homeFocusType>1</homeFocusType><br></config> |
|---------------------------------|-----------------------------------------------------------|

note:

focusType 0:manual    focus 1:auto focus

## 5.7.PTDC Zoom/Focus/Focus Type set information

### 5.7.1.set Zoom Mag

Set the Zoom absolute position in mag unit:

request:

GET/POST /cgi/ptdc.cgi?command=set\_zoom&zoom\_mag=5.5

parameters:

| Name     | Value | Description |
|----------|-------|-------------|
| zoom_mag | float | Zoom mag    |

response:

A result tag will also be presented:

|                               |                                          |                                  |
|-------------------------------|------------------------------------------|----------------------------------|
| Result:<br>Represented by XML | <result><br><code>ok</code><br></result> |                                  |
| Result code                   | 1. ok                                    | Success                          |
|                               | 2. invalidParameter                      | Need correct zoom_mag parameters |
|                               | 3. failure                               | Error command type               |
|                               | 4. oob                                   | Out Of Boundary                  |

### 5.7.2.set Relative Zoom Mag

Set the relative Zoom position in mag unit:

request:

GET/POST /cgi/ptdc.cgi?command=set\_relative\_zoom&zoom\_mag=2.5

parameters:

| Name     | Value | Description |
|----------|-------|-------------|
| zoom_mag | float | Zoom mag    |

response:

A result tag will also be presented:

|                               |                                          |                                  |
|-------------------------------|------------------------------------------|----------------------------------|
| Result:<br>Represented by XML | <result><br><code>ok</code><br></result> |                                  |
| Result code                   | 1. ok                                    | Success                          |
|                               | 2. invalidParameter                      | Need correct zoom_mag parameters |
|                               | 3. failure                               | Error command type               |
|                               | 4. oob                                   | Out Of Boundary                  |

Note:

Unit: mag

Add more 2.5x mag to current Zoom mag, ex original mag is 5.5x, and then goes to 8.0x after this command.

### 5.7.3.set Zoom Step

Set the absolute Zoom position in step unit:

request:

GET/POST /cgi/ptdc.cgi?command=set\_zoom\_step&zoom\_step=50

parameters:

| Name      | Value | Description |
|-----------|-------|-------------|
| zoom_step | #     | Zoom step   |

response:

A result tag will also be presented:

|                               |                                          |                                   |
|-------------------------------|------------------------------------------|-----------------------------------|
| Result:<br>Represented by XML | <result><br><code>ok</code><br></result> |                                   |
| Result code                   | 1. ok                                    | Success                           |
|                               | 2. invalidParameter                      | Need correct zoom_step parameters |
|                               | 3. failure                               | Error command type                |
|                               | 4. oob                                   | Out Of Boundary                   |

### 5.7.4.set Relative Zoom Step

Set the relative Zoom position in step unit:

request:

GET/POST /cgi/ptdc.cgi?command=set\_relative\_zoom\_step&zoom\_step=25

parameters:

| Name      | Value | Description |
|-----------|-------|-------------|
| zoom_step | #     | Zoom step   |

response:

A result tag will also be presented:

|                               |                                          |                                   |
|-------------------------------|------------------------------------------|-----------------------------------|
| Result:<br>Represented by XML | <result><br><code>ok</code><br></result> |                                   |
| Result code                   | 1. ok                                    | Success                           |
|                               | 2. invalidParameter                      | Need correct zoom_step parameters |
|                               | 3. failure                               | Error command type                |
|                               | 4. oob                                   | Out Of Boundary                   |

Note:

Unit: step

Add more 25 step to current Zoom step, ex original Zoom step is 50, and then goes to 75 after this command.

### 5.7.5.set Focus Step

Set the Focus absolute position in step unit:

request:

GET/POST /cgi/ptdc.cgi?command=set\_focus&focus\_step=500

parameters:

| Name       | Value | Description |
|------------|-------|-------------|
| focus_step | #     | Focus step  |

response:

A result tag will also be presented:

|                               |                                          |                                    |
|-------------------------------|------------------------------------------|------------------------------------|
| Result:<br>Represented by XML | <result><br><code>ok</code><br></result> |                                    |
| Result code                   | 1. ok                                    | Success                            |
|                               | 2. invalidParameter                      | Need correct focus_step parameters |
|                               | 3. failure                               | Error command type                 |
|                               | 4. oob                                   | Out Of Boundary                    |

## 5.7.6.set Relative Focus Step

Set the Focus relative position in step unit:

request:

GET/POST /cgi/ptdc.cgi?command=set\_relative\_focus\_step&focus\_step=25

parameters:

| Name       | Value | Description |
|------------|-------|-------------|
| focus_step | #     | Focus step  |

response:

A result tag will also be presented:

|                               |                                          |                                    |
|-------------------------------|------------------------------------------|------------------------------------|
| Result:<br>Represented by XML | <result><br><code>ok</code><br></result> |                                    |
| Result code                   | 1. ok                                    | Success                            |
|                               | 2. invalidParameter                      | Need correct focus_step parameters |
|                               | 3. failure                               | Error command type                 |
|                               | 4. oob                                   | Out Of Boundary                    |

Note:

Unit: step

Add more 50 step to current Focus step, ex original Focus step is 250, and then goes to 300 after this command.

## 5.7.7.set Focus Type

Set Focus type as manual or auto mode:

request:

GET/POST /cgi/ptdc.cgi?command=set\_focus\_type&focus\_type=0

parameters:

| Name       | Value  | Description |
|------------|--------|-------------|
| focus_type | 0 or 1 | Focus Type  |

response:

A result tag will also be presented:

|                               |                                          |                                   |
|-------------------------------|------------------------------------------|-----------------------------------|
| Result:<br>Represented by XML | <result><br><code>ok</code><br></result> |                                   |
| Result code                   | 1. ok                                    | Success                           |
|                               | 2. invalidParameter                      | Need correct focus_type parameter |
|                               | 3. failure                               | Error command type                |
|                               | 4. oob                                   | Out Of Boundary                   |

Note:

Focus\_type 0>manual focus 1:auto focus

## 5.8.Digital PTZ Control

### 5.8.1.get the current digital PTZ position

request:

GET /config/digital\_ptz\_pos.cgi

response:

| Name | Value | Description |
|------|-------|-------------|
|------|-------|-------------|

|           |            |                                                           |
|-----------|------------|-----------------------------------------------------------|
| profileid | #          | Profile number(# is a number from 1 to count of profiles) |
| p         | An integer | The digital pan position.                                 |
| t         | An integer | The digital tilt position.                                |
| z         | An integer | The digital zoom position.                                |

### 5.8.2.add, delete or goto a digital PTZ preset

request:

GET /config/digital\_ptz\_preset.cgi

parameters:

| Name      | Value        | Description                                               |
|-----------|--------------|-----------------------------------------------------------|
| profileid | #            | Profile number(# is a number from 1 to count of profiles) |
| act       | add, del, go | Action type.                                              |
| name      | A String     | Digital preset name                                       |

response:

see the above table. If the movement is out of boundary, you will get the actual absolute position.

### 5.8.3.move digital PTZ absolutely

request:

GET /config/digital\_ptz\_pos.cgi

parameters:

| Name      | Value      | Description                                               |
|-----------|------------|-----------------------------------------------------------|
| profileid | #          | Profile number(# is a number from 1 to count of profiles) |
| p         | An integer | Pans the device relative to the (0,0,0) position          |
| t         | An integer | Tilts the device relative to the (0,0,0) position         |
| z         | An integer | Zooms the device relative to the (0,0,0) position.        |

response:

see the above table. If the movement is out of boundary, you will get the actual absolute position.

### 5.8.4.move digital PTZ relatively

request:

GET /config/digital\_ptz\_move\_rel.cgi

parameters:

| Name      | Value      | Description                                               |
|-----------|------------|-----------------------------------------------------------|
| profileid | #          | Profile number(# is a number from 1 to count of profiles) |
| p         | An integer | Pans the device relative to the current position          |
| t         | An integer | Tilts the device relative to the current position         |
| z         | An integer | Zooms the device relative to the current position.        |

response:

see the above table. If the movement is out of boundary, you will get the actual relative p, t, z values it move.

### 5.8.5.digital PTZ autopan

request:

GET /config/digital\_ptz\_autopan.cgi

parameters:

| Name      | Value | Description                                               |
|-----------|-------|-----------------------------------------------------------|
| profileid | #     | Profile number(# is a number from 1 to count of profiles) |

|     |          |              |
|-----|----------|--------------|
| act | go, stop | Action type. |
|-----|----------|--------------|

response:

see the above table. If the movement is out of boundary, you will get the actual relative p, t, z values it move.

## 5.8.6.digital PTZ sequence

request:

GET /config/digital\_ptz\_sequence.cgi

parameters:

| Name      | Value              | Description                                               |
|-----------|--------------------|-----------------------------------------------------------|
| profileid | #                  | Profile number(# is a number from 1 to count of profiles) |
| act       | add, del, go, stop | Action type.                                              |
| name      | A string           | Only for “act=add”                                        |
| index     | An integer         | From 0 to max                                             |
| time      | An integer         | Only for “act=add”                                        |

response:

see the above table. If the movement is out of boundary, you will get the actual relative p, t, z values it move.

## 6.Notification API

---

### 6.1.Camera status notification

#### 6.1.1.get the notification status

This CGI is an one-shot command, which only return current status of IP camera. If you need to minitor camera status for a long time, please use 6.1.2. notify\_stream.cgi instead.

request:

GET /config/notify.cgi

response:

| Name          | Value                      | Description                                   |
|---------------|----------------------------|-----------------------------------------------|
| md#           | on, off                    | event motion detection # is triggered or not. |
| mdv#          | <degree of motion> (0-100) | Percentage of motion detected by camera.      |
| input#        | on, off                    | event input # is triggered or not.            |
| storagefull   | on, off                    | event storage full                            |
| storagefail   | on, off                    | event storage fail                            |
| recording     | on, off                    | status is recording                           |
| snaphooting   | on, off                    | status is snaphooting                         |
| output#       | on, off                    | status of output # is on or off               |
| vsignal       | on, off                    | status of video signal is on or lost          |
| speaker       | on, off                    | status of speaker is on or off                |
| mic           | on, off                    | status of microphone is on or off             |
| mdetecting    | on, off                    | setting of motion detection is on or off      |
| irled         | on, off                    | status of IR LED is on or off                 |
| autofocusbusy | yes, no                    | Status of autofocus is on or off              |

#### 6.1.2.get the notification stream

request:

GET /config/notify\_stream.cgi

response:

The client side should keep receiving notification information from camera. It includes all available events or status as follow table.  
The notification information is only generated on while event or status changed. If there is no any changed event or status being reported within 30 second, a special tag: "keep\_alive" will be send to the client side.

| Name        | Value                           | Description                                   |
|-------------|---------------------------------|-----------------------------------------------|
| md#         | on, off                         | event motion detection # is triggered or not. |
| mdv#        | <degree of motion> (0-100)      | Percentage of motion detected by camera.      |
| input#      | on, off                         | event input # is triggered or not.            |
| storagefull | on, off                         | event storage full                            |
| storagefail | on, off                         | event storage fail                            |
| recording   | on, off                         | status is recording                           |
| snaphooting | on, off                         | status is snaphooting                         |
| output#     | on, off                         | status of output # is on or off               |
| vsignal     | on, off                         | status of video signal is on or lost          |
| speaker     | on, off                         | status of speaker is on or off                |
| mic         | on, off                         | status of microphone is on or off             |
| usbstatus   | ready, protected, full, invalid | status of SD card inserted in the camera.     |

|               |               |                                  |
|---------------|---------------|----------------------------------|
| cameraname    | <camera name> | status of camera name            |
| irled         | on, off       | status of IR LED is on or off    |
| autofocusbusy | yes, no       | Status of autofocus is on or off |



## 7.RTSP API

---

The Real-Time Streaming Protocol (RTSP) is a protocol to get audio and video streaming data provided by a media server. IP camera can act as a media server and stream the real time audio and video . By RTSP request, a client application can get streaming data from IP camera. The detail of RTSP protocol please refer to RFC 2326.

### 7.1.Live streaming

#### 7.1.1.get URL entry of specified profile

request:

GET /config/rtspurl.cgi

parameters:

profileid=<video profile number>

response:

| Name      | Value                    | Description                                                    |
|-----------|--------------------------|----------------------------------------------------------------|
| profileid | #                        | profile number (# is a number from 1 to the count of profiles) |
| urlentry  | <entry of video profile> | URL entry of associated video stream profile                   |

#### 7.1.2.set video config

request:

GET/POST /config/rtspurl.cgi

parameters:

see the above table.

response:

see the above table.

#### 7.1.3.Get live video

The requested URI of an IP camera stream data can be described by following:

rtsp://<server ip>/<urlentry>

Get video and audio stream for use on PC.

Where <urlentry> is the url entry associated with one of the video profile. The value can be got by calling /config/rtspurl.cgi (see 7.1.1)

#### NOTE:

Since our camera now can let user to change the url entry of each video profile, the following RTSP urls are obsolete, user should use 7.1.3 to get RTSP stream.

rtsp://<server ip>/mp4

Get video and audio stream with MPEG-4 video format for use on PC.

rtsp://<server ip>/jpeg

Get video (and audio) stream with M-JPEG video format for use on PC.

rtsp://<server ip>/3gpp

Get video (and audio) stream with MPEG-4 video format for use on 3GPP compliant device.

rtsp://<server ip>/live#

where # is the number from 1 to the count of video profile. For example, use rtsp://192.168.1.1/live1 to get the stream of video profile number 1.

## 7.2 RTSP Methods:

A. OPTIONS: Report the methods supported by the IP camera.

Please use "OPTIONS" method to get the other methods supported by the IP camera.

## 8.Appendix

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### 8.1.Table used in NIPCA

*Table 1: Time zone*

| ID | Time zone                                                     |
|----|---------------------------------------------------------------|
| 1  | (GMT-12:00) International Date Line West                      |
| 2  | (GMT-11:00) Samoa                                             |
| 3  | (GMT-10:00) Hawaii                                            |
| 4  | (GMT-09:00) Alaska                                            |
| 5  | (GMT-08:00) Pacific Time (US & Canada)                        |
| 6  | (GMT-08:00) Tijuana, Baja California                          |
| 7  | (GMT-07:00) Chihuahua, La Paz, Mazatlan                       |
| 8  | (GMT-07:00) Mountain Time (US & Canada)                       |
| 9  | (GMT-07:00) Arizona                                           |
| 10 | (GMT-06:00) Central America                                   |
| 11 | (GMT-06:00) Guadalajara, Mexico City, Monterrey               |
| 12 | (GMT-06:00) Saskatchewan                                      |
| 13 | (GMT-06:00) Central Time (US & Canada)                        |
| 14 | (GMT-05:00) Bogota, Lima, Quito                               |
| 15 | (GMT-05:00) Eastern Time (US & Canada)                        |
| 16 | (GMT-05:00) Indiana (East)                                    |
| 17 | (GMT-04:00) La Paz, Georgetown                                |
| 18 | (GMT-04:00) Atlantic Time (Canada)                            |
| 19 | (GMT-04:00) Santiago                                          |
| 20 | (GMT-04:00) Manaus                                            |
| 21 | (GMT-03:30) Newfoundland                                      |
| 22 | (GMT-03:00) Buenos Aires                                      |
| 23 | (GMT-03:00) Brasilia                                          |
| 24 | (GMT-03:00) Greenland                                         |
| 25 | (GMT-03:00) Montevideo                                        |
| 26 | (GMT-02:00) Mid-Atlantic                                      |
| 27 | (GMT-01:00) Azores                                            |
| 28 | (GMT-01:00) Cape Verde Is.                                    |
| 29 | (GMT) Greenwich Mean Time : Dublin, Edinburgh, Lisbon, London |
| 30 | (GMT) Monrovia, Reykjavik                                     |
| 31 | (GMT+01:00) Belgrade, Bratislava, Budapest, Ljubljana, Prague |
| 32 | (GMT+01:00) West Central Africa                               |
| 33 | (GMT+01:00) Sarajevo, Skopje, Warsaw, Zagreb                  |
| 34 | (GMT+01:00) Brussels, Copenhagen, Madrid, Paris               |
| 35 | (GMT+01:00) Amsterdam, Berlin, Bern, Rome, Stockholm, Vienna  |
| 36 | (GMT+02:00) Helsinki, Kyiv, Riga, Sofia, Tallinn, Vilnius     |
| 37 | (GMT+02:00) Athens, Bucharest, Istanbul                       |
| 38 | (GMT+02:00) Beirut                                            |
| 39 | (GMT+02:00) Harare, Pretoria                                  |
| 40 | (GMT+02:00) Cairo                                             |
| 41 | (GMT+03:00) Minsk                                             |
| 42 | (GMT+02:00) Amman                                             |
| 43 | (GMT+01:00) Windhoek                                          |
| 44 | (GMT+02:00) Jerusalem                                         |
| 45 | (GMT+03:00) Baghdad                                           |

| <b>ID</b> | <b>Time zone</b>                                  |
|-----------|---------------------------------------------------|
| 46        | (GMT+03:00) Moscow, St. Petersburg, Volgograd     |
| 47        | (GMT+04:00) Tbilisi                               |
| 48        | (GMT+03:00) Nairobi                               |
| 49        | (GMT+03:00) Kuwait, Riyadh                        |
| 50        | (GMT+03:30) Tehran                                |
| 51        | (GMT+04:00) Baku                                  |
| 52        | (GMT+04:00) Abu Dhabi, Muscat                     |
| 53        | (GMT+04:00) Yerevan                               |
| 54        | (GMT+04:30) Kabul                                 |
| 55        | (GMT+06:00) Yekaterinburg                         |
| 56        | (GMT+05:00) Islamabad, Karachi, Tashkent          |
| 57        | (GMT+05:30) Chennai, Kolkata, Mumbai, New Delhi   |
| 58        | (GMT+05:30) Sri Jayawardenepura                   |
| 59        | (GMT+05:45) Kathmandu                             |
| 60        | (GMT+06:00) Astana, Dhaka                         |
| 61        | (GMT+07:00) Novosibirsk                           |
| 62        | (GMT+06:30) Yangon (Rangoon)                      |
| 63        | (GMT+08:00) Krasnoyarsk                           |
| 64        | (GMT+07:00) Bangkok, Hanoi, Jakarta               |
| 65        | (GMT+08:00) Beijing, Chongqing, Hong Kong, Urumqi |
| 66        | (GMT+08:00) Taipei                                |
| 67        | (GMT+09:00) Irkutsk, Ulaan Bataar                 |
| 68        | (GMT+08:00) Perth                                 |
| 69        | (GMT+08:00) Kuala Lumpur, Singapore               |
| 70        | (GMT+10:00) Yakutsk                               |
| 71        | (GMT+09:00) Osaka, Sapporo, Tokyo                 |
| 72        | (GMT+09:00) Seoul                                 |
| 73        | (GMT+09:30) Adelaide                              |
| 74        | (GMT+09:30) Darwin                                |
| 75        | (GMT+10:00) Hobart                                |
| 76        | (GMT+10:00) Brisbane                              |
| 77        | (GMT+10:00) Vladivostok                           |
| 78        | (GMT+10:00) Canberra, Melbourne, Sydney           |
| 79        | (GMT+10:00) Guam, Port Moresby                    |
| 80        | (GMT+12:00) Magadan                               |
| 81        | (GMT+12:00) Fiji                                  |
| 82        | (GMT+12:00) Auckland, Wellington                  |
| 83        | (GMT+13:00) Nukualofa                             |
| 84        | (GMT-04:30) Caracas                               |
| 85        | (GMT+11:00) Solomon Is., New Caledonia            |
| 86        | (GMT) Casablanca                                  |
| 87        | (GMT+08:00) Ulaanbaatar                           |

*Table 2: dynamic DNS service providers*

| <b>ID</b> | <b>provider URIs</b>   |
|-----------|------------------------|
|           | www.ez-ip.net          |
|           | www.penguinpowered.com |
|           | members.dhs.org        |
| dyndns    | members.dyndns.org     |
|           | www.3322.org           |
|           | update.ods.org         |
|           | cgi.tzo.com            |
|           | members.easydns.com    |
|           | api.easydns.com        |

|  |                   |
|--|-------------------|
|  | www.justlinux.com |
|  | www.dyns.cx       |
|  | dup.hn.org        |
|  | www.zoneedit.com  |
|  | ipv6tb.he.net     |

## 8.2.Advanced IP-Camera Stream (ACS) Header

*Multimedia header:*

| ACS Audio header                                                                                                                                                                                                                                                                                                                                                                                                                     | ACS Video header                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre>typedef struct _ACS_AudioHeader { unsigned long ulHdrID; //Header ID unsigned long ulHdrLength; unsigned long ulDataLength; unsigned long ulSequenceNumber; unsigned long ulTimeSec; unsigned long ulTimeUSec; unsigned long ulDataChecksum; unsigned short usFormat; unsigned short usChannels; unsigned short usSampleRate; unsigned short usSampleBits; unsigned long ulReserved; }ACS_AudioHeader, *PACS_AudioHeader;</pre> | <pre>typedef struct _ACS_VideoHeader { unsigned long ulHdrID; //Header ID unsigned long ulHdrLength; unsigned long ulDataLength; unsigned long ulSequenceNumber; unsigned long ulTimeSec; unsigned long ulTimeUSec; unsigned long ulDataChecksum; unsigned short usCodingType; unsigned short usFrameRate; unsigned short usWidth; unsigned short usHeight; unsigned char ucMDBitmap; unsigned char ucMDPowers[3]; }ACS_VideoHeader, *PACS_VideoHeader</pre> |

Description:

The byte order of this header is little-endian.

Common part:

**ulHdrID:** Special id for identifying ACS header. For audio: the value of this id is 0xF6010000 (since our header is in little-endian so the byte array of this id is '00 00 01 F6'). For video the value is 0xF5010000.

**ulHdrLength:** Header length. (32 bytes in current version)

**ulDataLength:** Payload data length.

**ulSequenceNumber:** Sequence number.

**ulTimeSec:** Time stamp in sec since 1970/01/01 00:00.

**ulTimeUSec:** Microsecond part of time stamp

**ulDataChecksum:** Store last 4 bytes of payload data.

Audio part:

**usFormat:** Audio data format. The possible value:

AFMT\_MS\_ADPCM: 0

AFMT\_MU\_LAW: 1

AFMT\_A\_LAW: 2

AFMT\_IMA\_ADPCM: 4

AFMT\_U8: 8

AFMT\_S16\_LE: 0x10 /\* Little endian signed 16\*/

AFMT\_S16\_BE: 0x20 /\* Big endian signed 16 \*/

AFMT\_S8: 0x40

AFMT\_U16\_LE: 0x80 /\* Little endian U16 \*/

AFMT\_U16\_BE: 0x100 /\* Big endian U16 \*/

AFMT\_MPEG: 0x200 /\* MPEG (2) audio \*/

AFMT\_AC3: 0x400

AFMT\_AMR: 0x800

AFMT\_AAC: 0x1000  
 AFMT\_ALAW: 0x2000  
 usChannels: Audio channels number: mono(1) or stereo(2).  
 usSampleRate: Sample rate.  
 usSampleBits: Bits count per sample.  
 ulReseverd: Reserved.

#### Video part:

usCodingType: Encoding type of frame. The possible values are:

VFCT\_IVOP (MPEG4): 0  
 VFCT\_PVOP (MPEG4): 1  
 VFCT\_JPEG: 5  
 VFCT\_H264\_IFRM: 10  
 VFCT\_H264\_PFRM: 11

usFrameRate: Frames per second.  
 usWidth: The width of frame dimension  
 usHeight: The height of frame dimension  
 ucMDBitmap: The height of frame dimension  
 ucMDPowers[3]: The height of frame dimension

#### *Extension header:*

We propose add extensive header for dealing with other information attaching with the multimedia stream. Instead of appending this kind of information to multimedia stream, it can save more bandwidth utilization.

Table: Alphanetworks IP-Camera streaming (ACS) extension header:

| ACS extension header                                                                                                                                                                         |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre>typedef struct _ACS_ExtentHeader {   unsigned long ulHdrID; // '00 00 01 FE'   unsigned long ulHdrLength;   unsigned char pbyReserved[96]; } ACS_ExtentHeader, *PACS_ExtentHeader</pre> |

#### Description:

The extension header is interleaved within the video stream or audio stream when the information is required by client.

ulHdrID: Special id for identifying ACS header. 0xFE010000.  
 ulHdrLength: Header length. (32 bytes in current version)  
 pbyReserved[96]: To be defined.